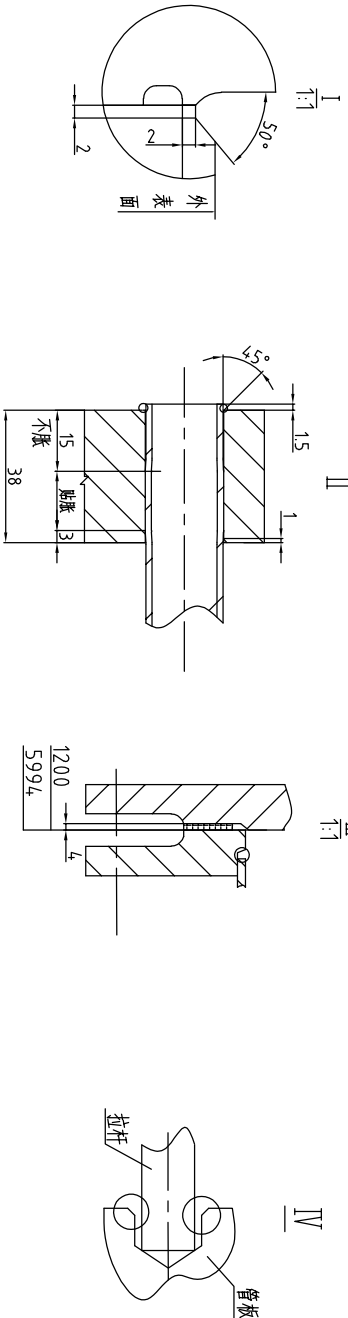
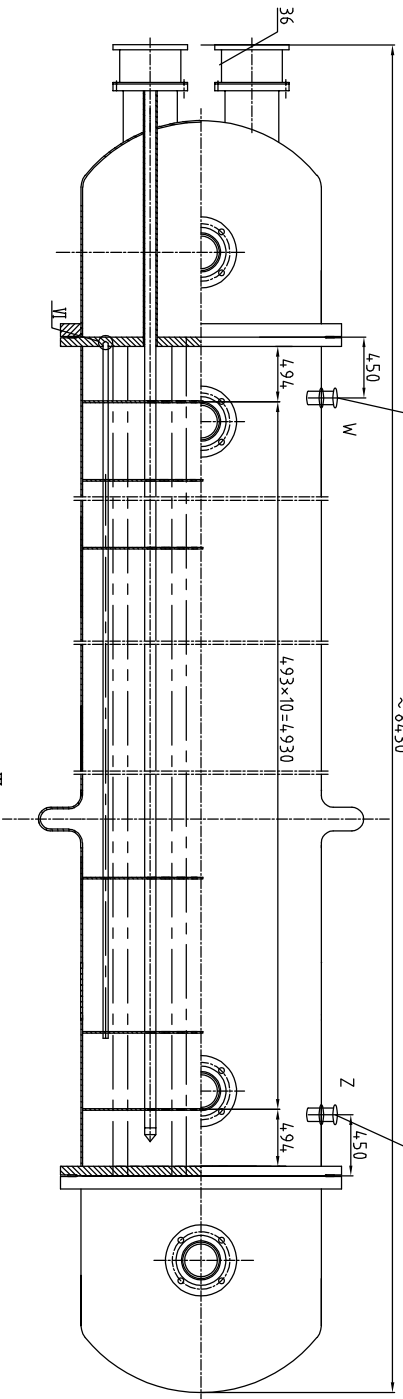
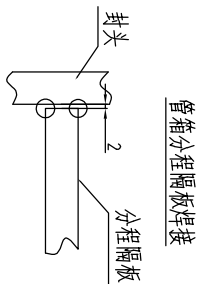
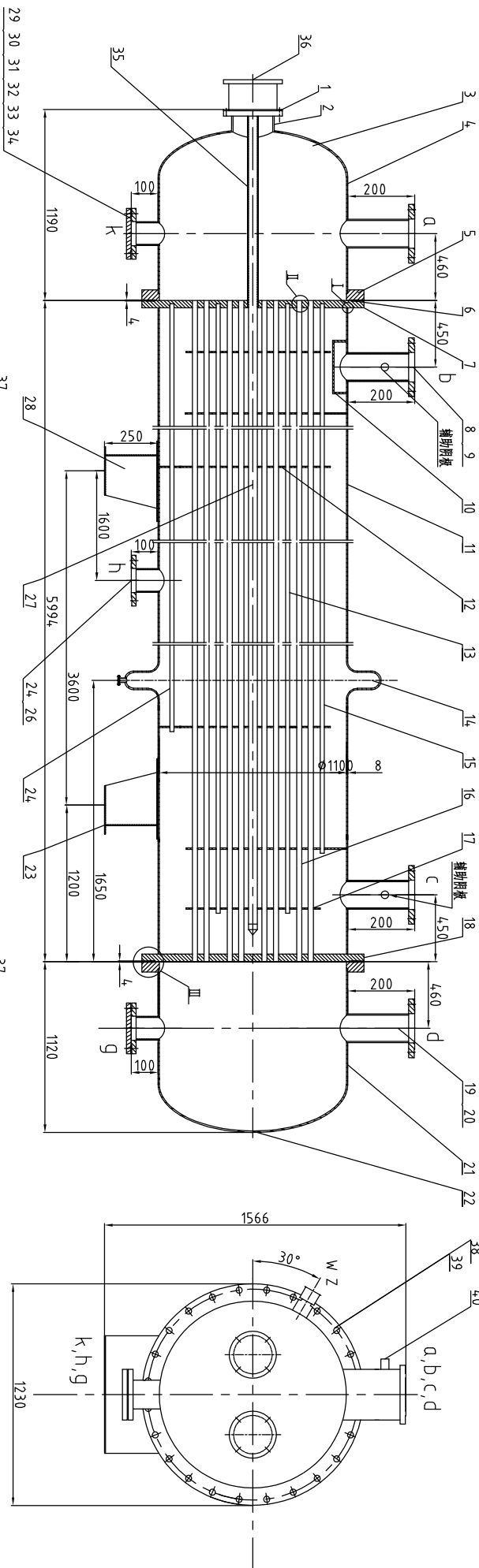


技术要求

- 阳极保护酸冷器设计、制造、质量控制和验收参照GB/T150.1~150.4-2024《压力容器》和GB/T151-2014《热交换器》标准。
- 阳极保护酸冷器制造生产管理和工艺控制以及质量检验遵照企业标准LZXYY.GL.001-2011《生产过程控制程序》、LZXYY.GL.002-2011《酸冷器制造工艺和质量检验守则》。
- 焊接坡口型式依据GB/T985.1-2008《气焊、焊条电弧焊、气体保护焊和高能束焊的推荐坡口》选取，坡口加工采用机械方法制作或热切割后再进行砂轮机加工修整。
- 酸冷器焊缝表面不得有裂纹、未焊透、未熔合、气孔、弧坑、夹渣以及焊缝两侧飞溅等；不锈钢焊接接头不得有交边。
- 依据GB/T150.1~150.4-2024《压力容器》对管、壳程进行水压试验，试验压力为1.25倍的设计压力。
- 设备制造完毕后，不锈钢表面使用不含盐酸的酸洗钝化膏进行酸洗钝化处理。
- 油漆、包装、运输按NB/T10558-2021《压力容器涂敷与运输包装》的规定。



技术特性表

名 称	单 位	管 程	壳 程
容器类别		管壳式换热器	
设计压力	Mpa	0.5	0.6
设计温度	℃	75	120
工作压力	Mpa	0.4	0.5
工作温度	℃	32-40	92.5-65
物料名称		循环水	硫酸 98.5%
物料用量	Kg/h	14.32592	114.74.00
换热面积	M ²	4.05	
传热系数	w/(m ² ℃)	672.7	
焊缝系数		0.85	0.85
操作重量	kg	26000	

管口表

符 号	工 程 尺 寸	链接尺寸标准	连接面形式	用途或名称
a	500	PL500(B)-10,HG/T20592	RF	出水口
b	500	S0500(B)-16,HG/T20592	RF	进酸口
c	500	S0500(B)-16,HG/T20592	RF	出酸口
d	500	PL500(B)-10,HG/T20592	RF	进口水
g	80	PL80(B)-10,HG/T20592	RF	排污口
h	80	S080(B)-16,HG/T20592	RF	排污口
k	80	PL80(B)-10,HG/T20592	RF	排污口

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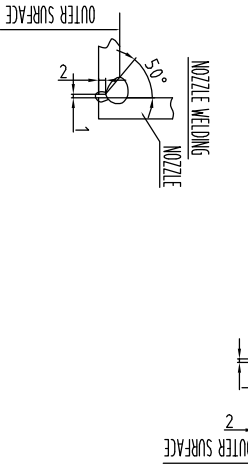
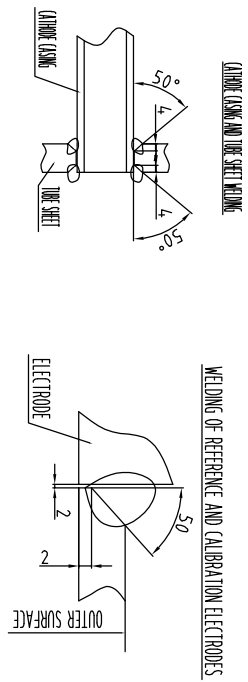
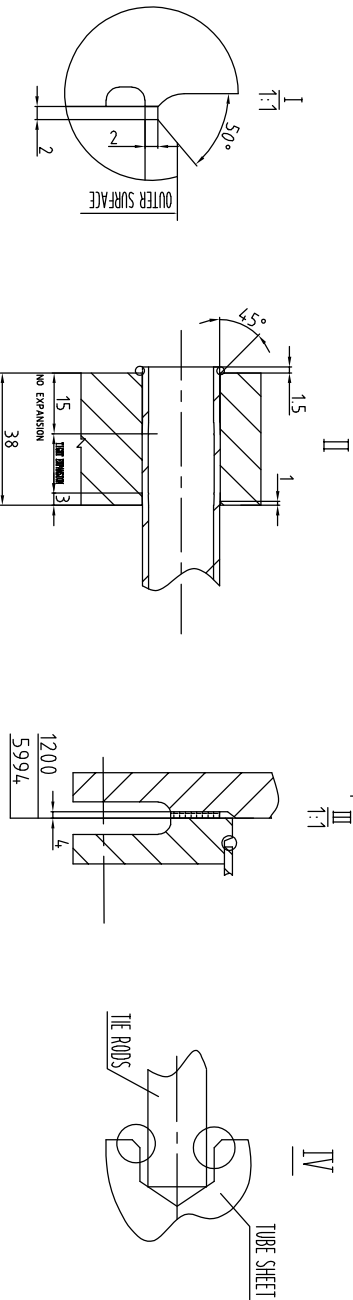
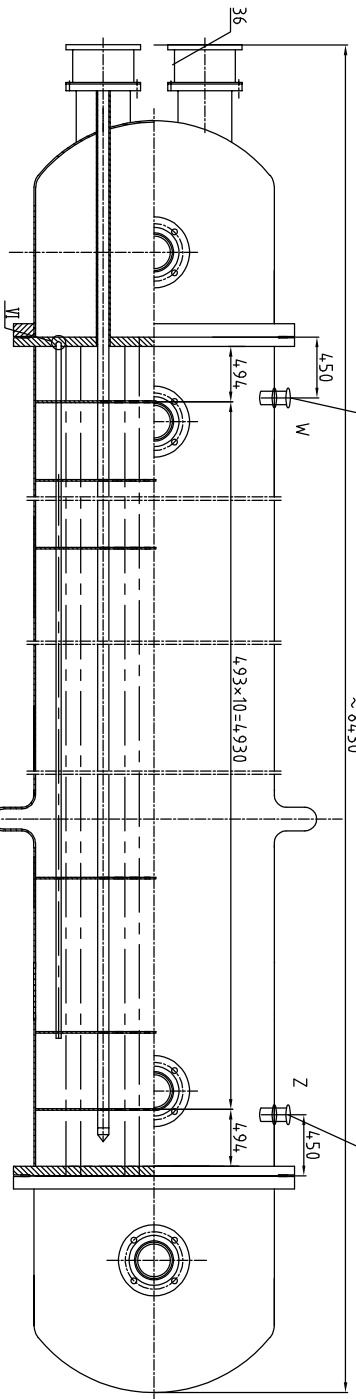
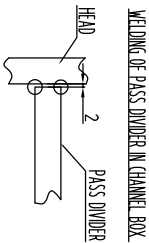
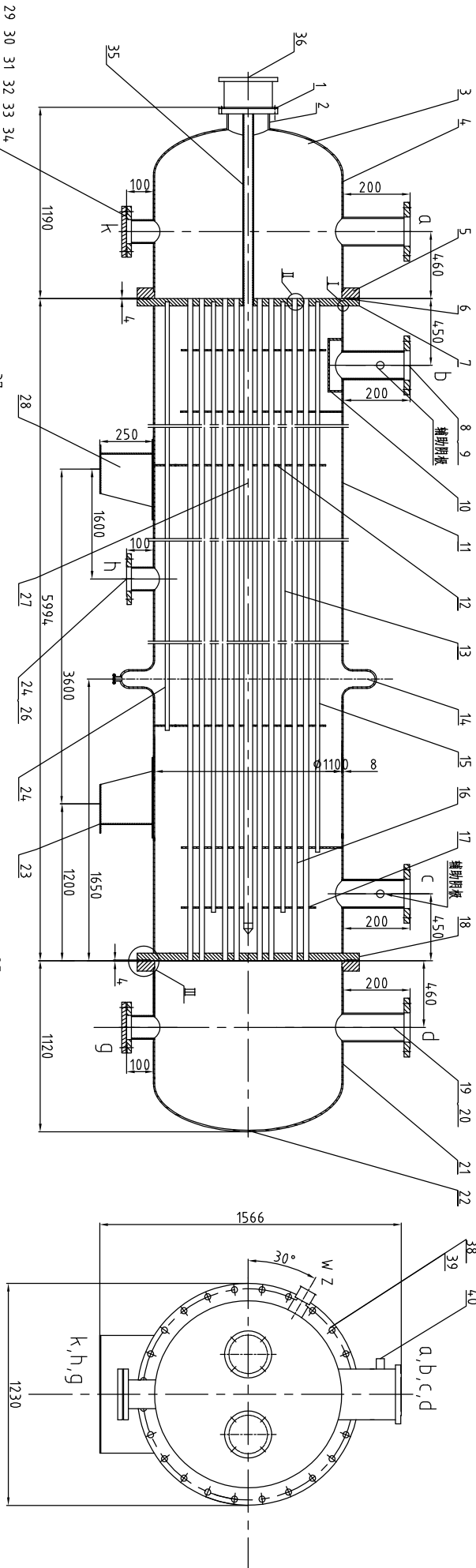
设计		位号	EX-11301
制图		图号	HR2508-0
审核		比例	重量Kg 8450
审定			

干燥酸冷却器

马兹别克斯坦年产50万吨硫酸制酸项目

TECHNICAL REQUIREMENTS

- The design, manufacture, quality control and acceptance of anodic protection acid cooler refer to the standards of GB/T150.1-150.4-2024, "Pressure Vessel" and GB/T151-2014, "Heat Exchanger".
- Anode protection acid cooler manufacturing production management and process control and quality inspection in accordance with the enterprise standard LZXY.GI.001-2011 "Production Process Control Procedures".
- LZXY.GI.002-2011 "Acid cooler manufacturing process and Quality Inspection Code"
- The welding groove type shall be selected in accordance with GB/T 985.1 - 2008 Recommended grooves for oxyfuel gas welding, Shielded Metal Arc Welding, Gas-Shielded Welding, and High-Energy Beam Welding for groove processing, a mechanical method shall be used for fabrication or after thermal cutting. it shall be framed by grinding wheel machining.
- The weld surface of the acid cooler must not have cracks, incomplete penetration, lack of fusion, pores, arc craters, slag inclusions, and spatter on both sides of the weld. For stainless steel welded joints, there must be no undercut.
- In accordance with GB/T 150.1 - 150.4 - 2024, Pressure Vessel hydrostatic tests shall be conducted on the tubes and the shell side. The test pressure shall be 1.25 times the design pressure.
- After the equipment is manufactured the stainless steel surface shall be subjected to pickling and passivation treatment using a pickling and passivating paste that does not contain hydrochloric acid.
- Painting, packaging, transportation according to NB/T10558-2021 "Pressure Vessel coating and transportation packaging" provisions.



TECHNICAL CHARACTERISTIC TABLE

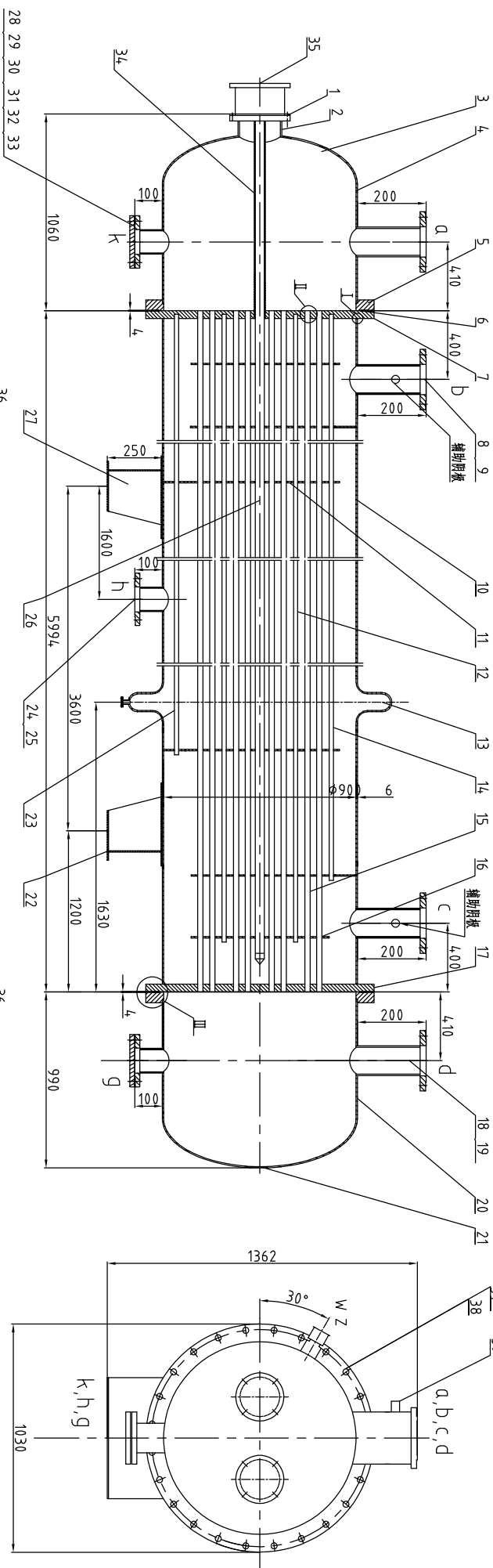
NAME	UNIT	TUBE SIDE	SHELL SIDE
TYPE OF VESSEL		SHELL AND TUBE	HEAT EXCHANGER
DESIGN PRESS	Mpa	0.5	0.6
DESIGN TEMP	℃	75	120
OPERATING PRESS	Mpa	0.4	0.5
OPERATING TEMP	℃	32-40	92.5-65
MEDIUM NAME		Circulation Water	H2SO4, 98.5%
MEDIUM DOSAGE	Kg/h	14.32592	114.71400
HEAT EXCHANGE AREA	M ²		4.05
HEAT TRANSFER COEFFICIENT	W/(m ² ℃)		672.7
UNIT EFFICIENCY		0.85	0.85
OPERATION WEIGHT	kg		26000

NOZZLE SCHEDULE

MARK	NOMINAL SIZE	CONNECTION STANDARD	CONNECTION SURFACE FORM	USER OR NAME
a	500	PL500(B)-10, HG/T20592	RF	Water Outlet
b	500	S0500(B)-16, HG/T20592	RF	Acid Inlet
c	500	S0500(B)-16, HG/T20592	RF	Acid Outlet
d	500	PL500(B)-10, HG/T20592	RF	Water Inlet
g	80	PL80(B)-10, HG/T20592	RF	Sewage outlets
h	80	S080(B)-16, HG/T20592	RF	Sewage outlets
k	80	PL80(B)-10, HG/T20592	RF	Sewage outlets

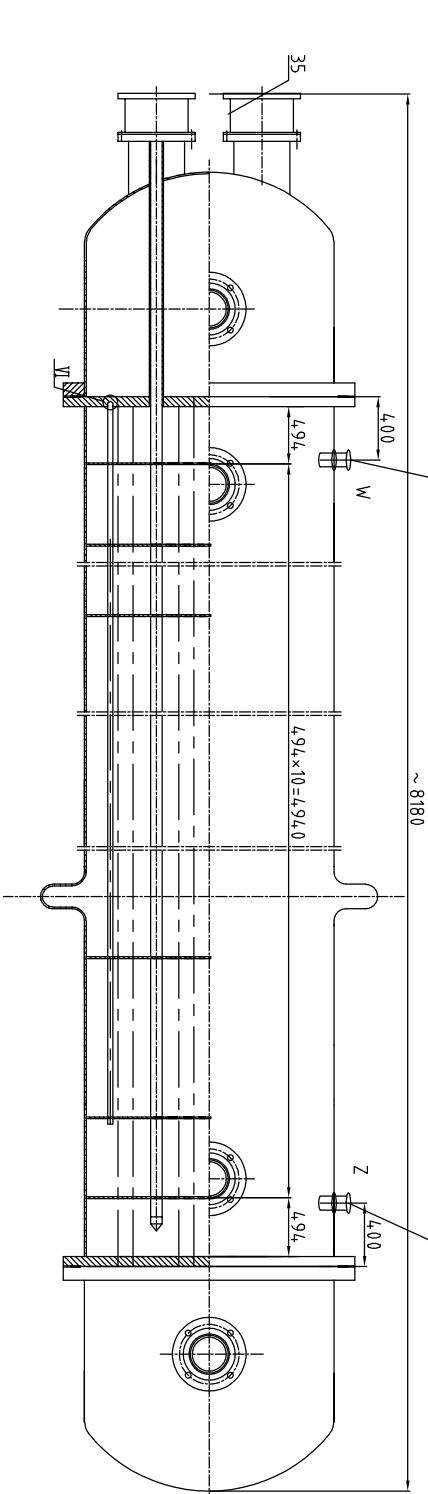
40	auxiliary cathode	ASSEMBLY	2		
39	GB6170 NUTS M20	Q235B	88		
38	GB5782 BOLTS M20×150	Q235B	88		
37	REFERENCE ELECTRODE	ASSEMBLY	2		
36	MAIN CATHODE	ASSEMBLY	2		
35	CATHODE CASING Ø40×3.5, L=1220	316L	2		
34	GB6170 NUTS M16	Q235B	16		
33	GB5782 BOLTS M16×65	Q235B	16		
32	HG/T20592 FLANGE COVER BL80-10	Q235B	2		
31	HG/T20592 GASKETS RF80-10, Ø=2	橡胶板	2		
30	HG/T20592 PIPE FLANGE PL80(B)-10, RF	Q235B	2		
29	NOZZLE Ø89×4	20	2		
28	JB/T4712.1-2007 LEFT SUPPORT B11100-F, H=250	Q235B	1		
27	CATHODE FRONT Ø201.6, 6995 XX-1 ALLOY	2			
26	HG/T20592-2009 PIPE FLANGE S080(B)-16, RF	304L	1		
25	NOZZLE Ø89×4	304L	1		
24	Short pull rod Ø14, L=3975	304L	1		
23	JB/T4712.1-2007 RIGHT SUPPORT B11100-S, H=250	Q235B	1		
22	GB/T25198-2010 ELLIPTICAL HEAD DN1100×10	Q235B	1		
21	RIGHT CHANNEL NOZZLE STUB DN1100×10, L=712	Q235B	1		
20	HG/T20592-2009 PIPE FLANGE PL500(B)-10, RF	Q235B	2		
19	NOZZLE Ø530×10	20	2		
18	HR2508-3 RIGHT TUBE SHEET Ø=38	304L	1		
17	HR2508-4 circular base slab Ø=10	304L	6		
16	HEAT EXCHANGER TUBES Φ19×1.5, L=6000	316L	114.5		
15	centre pull rod Ø14, L=4961	304L	1		
14	GB/T 16719 EXPANSION JOINT DN1100, Ø=8	304L	1		
13	Long pull rod Ø14, L=5454	304L	8		
12	HR2508-2 B A F F L E Ø=10	304L	5		
11	SHELL SIDE CYLINDER DN1100×8, L=5918	304L	1		
10	Strike plate Ø560 Ø=10	304L	1		
9	HG/T20592-2009 PIPE FLANGES0500(B)-16, RF	304L	2		
8	NOZZLE Ø530×10	304L	2		
7	HR2508-1 LEFT TUBE SHEET Ø=38	304L	1		
6	GASKETS DN1100 PNO.6, Ø=4	RUBBER SHEET	2		
5	NB/T 47021 FLANGE DN1100, PNO.6	Q235B	2		
4	LEFT CHANNEL NOZZLE STUB DN1100×10, L=712	Q235B	1		
3	GB/T25198-2010 ELLIPTICAL HEAD DN1100×10	Q235B	1		
2	NOZZLE Ø159×5	20	2		
1	CATHODE HOLDER Ø220, Ø=20	Q235B	2		

Part No.	NAME	MATERIAL	QTY	UNIT	TOTAL WT (KG)	REMARKS
Lanzhou Xingye Material Protection Co., Ltd. 2025.12						
DESIGNED		Position	number	EX-11301		
DRAWING		Anodically protected drying acid cooler		DRAWING NO.	HR2508-0	
CHECKED				REVIEWED		
APPROVED		The 500,000-ton sulfur-based acid production project in Uzbekistan		SCALE	WEIGHT (kg)	8450



技术要求

- 阳极保护酸冷器设计、制造、质量控制和验收参照GB/T150.1~150.4-2024《压力容器》和GB/T151-2014《热交换器》标准。
- 阳极保护酸冷器制造生产管理和工艺控制以及质量检验遵照企业标准LZXY·GL·001-2011《生产过程控制程序》、LZXY·GL·002-2011《酸冷器制造工艺和质量检验守则》。
- 焊接坡口型式依据GB/T985.1-2008《气焊、焊条电弧焊、气体保护焊和高能束焊的推荐坡口》选取，坡口加工采用机械方法制作或热切割后再进行砂轮磨加工修整。
- 酸冷器焊缝表面不得有裂纹、未焊透、未熔合、气孔、弧坑、夹渣以及焊缝两侧飞溅等；不锈钢焊接接头不得有咬边。
- 依据GB/T150.1~150.4-2024《压力容器》对管、壳程进行水压试验，试验压力为1.25倍的设计压力。
- 设备制造完毕后，不锈钢表面使用不含盐酸的酸洗钝化膏进行酸洗钝化处理。
- 油漆、包装、运输按NB/T10558-2021《压力容器涂装与运输包装》的规定。

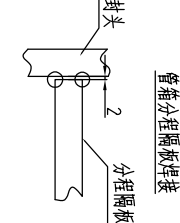


技术特性表

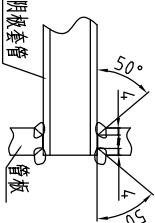
名称	单位	管程	壳程
容器类别		管壳式换热器	
设计压力	MPa	0.5	0.6
设计温度	℃	75	120
工作压力	MPa	0.4	0.5
工作温度	℃	32~4.0	92.5~71
物料名称		循环水	硫酸 98.5%
物料用量	Kg/h	1004852	1025600
换热面积	M ²		260
传热系数	W/(m ² ·℃)		680.4
焊接系数		0.85	0.85
操作重量	kg		17000

管口表

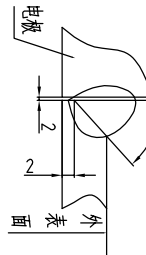
符号	工程尺寸	链接尺寸标准	连接面形式	用途或名称
a	4.50	PL4.50(B)-10.HG/T20592	RF	出水口
b	4.50	SO4.50(B)-16.HG/T20592	RF	进酸口
c	4.50	SO4.50(B)-16.HG/T20592	RF	出酸口
d	4.50	PL4.50(B)-10.HG/T20592	RF	进水口
g	80	PL80(B)-10.HG/T20592	RF	排污口
h	80	SO80(B)-16.HG/T20592	RF	排污口
k	80	PL80(B)-10.HG/T20592	RF	排污口



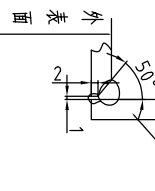
隔板套管与管板焊接



对比及校正电极焊接



接管焊接



作号	图号或标准	名	称	材	料	数量	重量	重量	备
1		接管	φ159×5	20	2				
2		隔板套	φ220, δ=20	Q235B	2				
3	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
4	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
5	NB/T 47021	管法兰	PN0.6, δ=4	Q235B	2				
6		垫片	DN900, PNO.6	橡胶板	2				
7	HR2509-1	左管板	φ4,80×10	304L	1				
8	HR2509-1	右管板	φ4,80×10	304L	1				
9	HG/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
10	HG/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
11	HR2509-2	左管板	φ4,80×10	304L	1				
12	HR2509-2	右管板	φ4,80×10	304L	1				
13	GB/T 16749	中拉杆	φ14, L=4970	304L	1				
14	GB/T 16749	中拉杆	φ14, L=5464	304L	8				
15	HR2509-4	左管板	φ4,80×10	304L	1				
16	HR2509-4	右管板	φ4,80×10	304L	1				
17	HR2509-3	左管板	φ4,80×10	304L	1				
18	HR2509-3	右管板	φ4,80×10	304L	1				
19	HG/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
20	HG/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
21	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
22	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
23	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
24	HR2509-4	左管板	φ4,80×10	304L	1				
25	HR2509-4	右管板	φ4,80×10	304L	1				
26	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
27	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
28	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
29	GB/T20592-2008	管法兰	PN0.6, δ=4	Q235B	1				
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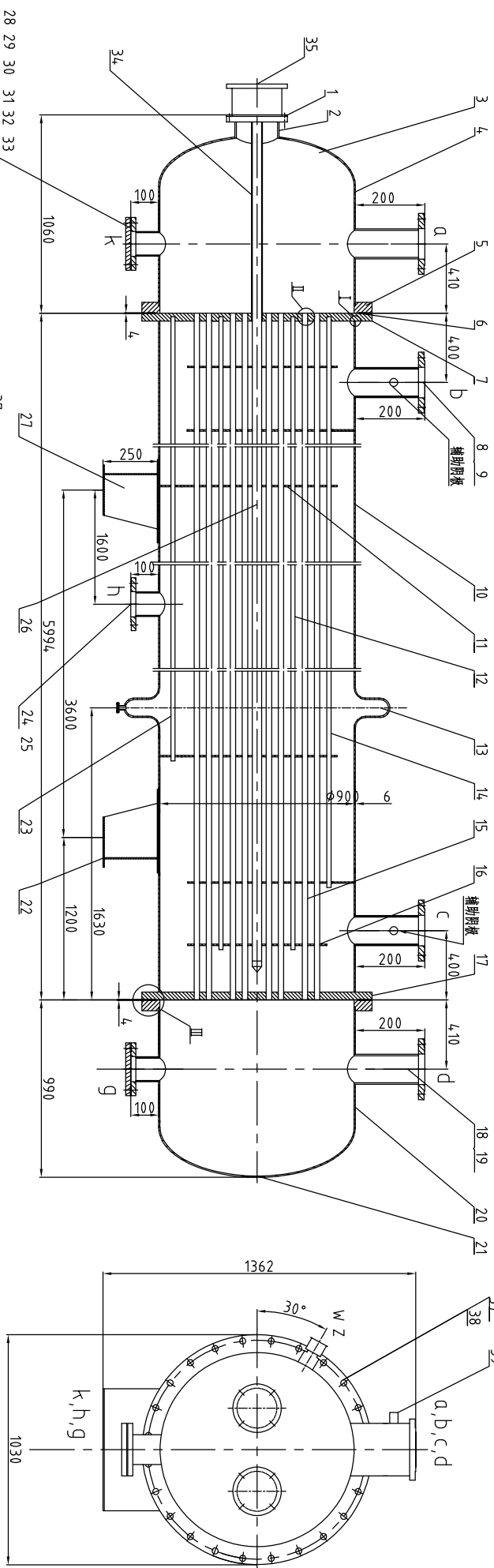
一吸酸冷器

HR2509-0

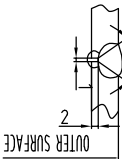
重量kg 5420

TECHNICAL REQUIREMENTS

- 1.The design, manufacture, quality control and acceptance of anodic protection acid cooler refer to the standards of GB/T150.1-150.4-2024 "Pressure Vessel" and GB/T151-2014 "Heat Exchanger".
- 2.Anodic protection acid cooler manufacturing production management and process control and quality inspection in accordance with the enterprise standard LZXY GL 001-2011 "Production Process Control Procedures".
- 3.The welding groove type shall be selected in accordance with GB/T 985. 1 - 2008 LZXY GL 002-2011 "Acid cooler manufacturing process and Quality Inspection Code"
4. The weld surface of the acid cooler must not have cracks, incomplete penetration, lack of fusion, pores, arc craters, slag inclusions, and spatter on both sides of the weld. For stainless steel welded joints, there must be no undercuts.
5. In accordance with GB/T 150.1 - 150.4 - 2024, Pressure Vessel,hydrostatic tests shall be conducted on the tubes and the shell side. The test pressure shall be 1.25 times the design pressure.
6. After the equipment is manufactured, the stainless steel surface shall be subjected to pickling and passivation treatment using a pickling and passivating paste that does not contain hydrochloric acid.
- 7.Painting, packaging, transportation according to NB/T10558-2021 "Pressure Vessel coating and transportation packaging" provisions.



LONGITUDINAL AND CIRCUMFERENTIAL WELDS OF SHELL



NOZZLE SCHEDULE

MARK	NOMINAL SIZE	CONNECTION STANDARD	CONNECTION SURFACE FORM	USER OR NAME
a	4.50	PL450(B)-10.HG/T20592	RF	Water Outlet
b	4.50	SO450(B)-16.HG/T20592	RF	Acid Inlet
c	4.50	SO450(B)-16.HG/T20592	RF	Acid Outlet
d	4.50	PL450(B)-10.HG/T20592	RF	Water Inlet
g	80	PL80(B)-10.HG/T20592	RF	Sewage outlets
h	80	SO80(B)-16.HG/T20592	RF	Sewage outlets
k	80	PL80(B)-10.HG/T20592	RF	Sewage outlets

TECHNICAL CHARACTERISTIC TABLE

NAME	UNIT	TUBE SIDE	SHELL SIDE
TYPE OF VESSEL		SHELL AND TUBE	HEAT EXCHANGER
DESIGN PRESS	Mpa	0.5	0.6
DESIGN TEMP	℃	75	120
OPERATING PRESS	Mpa	0.4	0.5
OPERATING TEMP	℃	32-40	92.5-71
MEDIUM NAME		Circulation Water	H2SO4, 98.5%
MEDIUM DOSAGE	Kg/h	1004852	1025600
HEAT EXCHANGER	M ²		260
HEAT TRANSFER	W/(m ² ·℃)		680.4
LOW EFFICIENCY		0.85	0.85
OPERATION WEIGHT	kg		17000

39	auxiliary cathode	ASSEMBLY	2		
38	GB6170	NUTS M20	Q235B	64	
37	GB5782	BOLTS M20×120	Q235B	64	
36		REFERENCE ELECTRODE	ASSEMBLY	2	
35		MAIN CATHODE	ASSEMBLY	2	
34		CATHODE CASING φ40×35L=1085	316L	2	
33	GB6170	NUTS M16	Q235B	16	
32	GB5782	BOLTS M16×65	Q235B	16	
31	HG/T20592	FLANGE COVER BL80-10	Q235B	2	
30	HG/T20592	GASKETS RF80-10,δ=2	RUBBER SHEET	2	
29	HG/T20592	PIPE FLANGE PL80(B)-10,RF	Q235B	2	
28		NOZZLE φ89×4	20	2	
27	JB/T4712.1-2007	LEFT SUPPORT BL1900-F.H=250	Q235B	1	
26		CATHODE FRONT φ20.L=665XY-1	ALLOY	2	
25	HG/T20592-2009	PIPE FLANGE SO80(B)-16,RF	304L	1	
24		NOZZLE φ89×4	304L	1	
23		hot pull rod φ14,L=4075	304L	1	
22	JB/T4712.1-2007	RIGHT SUPPORT BL1900-S.H=250	Q235B	1	
21	GB/T2598-2008	ELLIPTICAL HEAD DN900×8	Q235B	1	
20		RIGHT CHAMF. NOZZLE SUB DN900×10	Q235B	1	
19	HG/T20592-2009	PIPE FLANGE PL450(B)-10,RF	Q235B	2	
18		NOZZLE φ480×10	20	2	
17	HR2509-3	RIGHT TUBE SHEET δ=33	304L	1	
16	HR2509-4	circular base slab δ=8	304L	6	
15		HEAT EXCHANGER TUBES φ8×51-4MM	316L	735	
14		centre pull rod φ14,L=4970	304L	1	
13	GB/T 16749	EXPANSION JOINT DN900,δ=6	304L	1	
12		Long pull rod φ14,L=5464	304L	8	
11	HR2509-2	BAFFLE δ=8	304L	5	
10		SHELL SIDE CYLINDER DN900×1.570	304L	1	
9	HG/T20592-2009	PIPE FLANGE SO450(B)-16,RF	304L	2	
8		NOZZLE φ480×10	304L	2	
7	HR2509-1	LEFT TUBE SHEETδ=33	304L	1	
6		GASKETS DN900 PNO 6,δ=4	RUBBER SHEET	2	
5	NB/T 47021	FLANGE DN900,PNO 6	Q235B	2	
4		LEFT CHAMF. NOZZLE SUB DN900×10	Q235B	1	
3	GB/T2598-2008	ELLIPTICAL HEAD DN900×8	Q235B	1	
2		NOZZLE φ159×5	20	2	
1	HR2509-4	CATHODE HOLDER φ220,δ=20	Q235B	6	
PART NO.	NAME	MATERIAL	QT	UNIT	TOT
DN900 XINGYE				WT(KG)	REMARKS

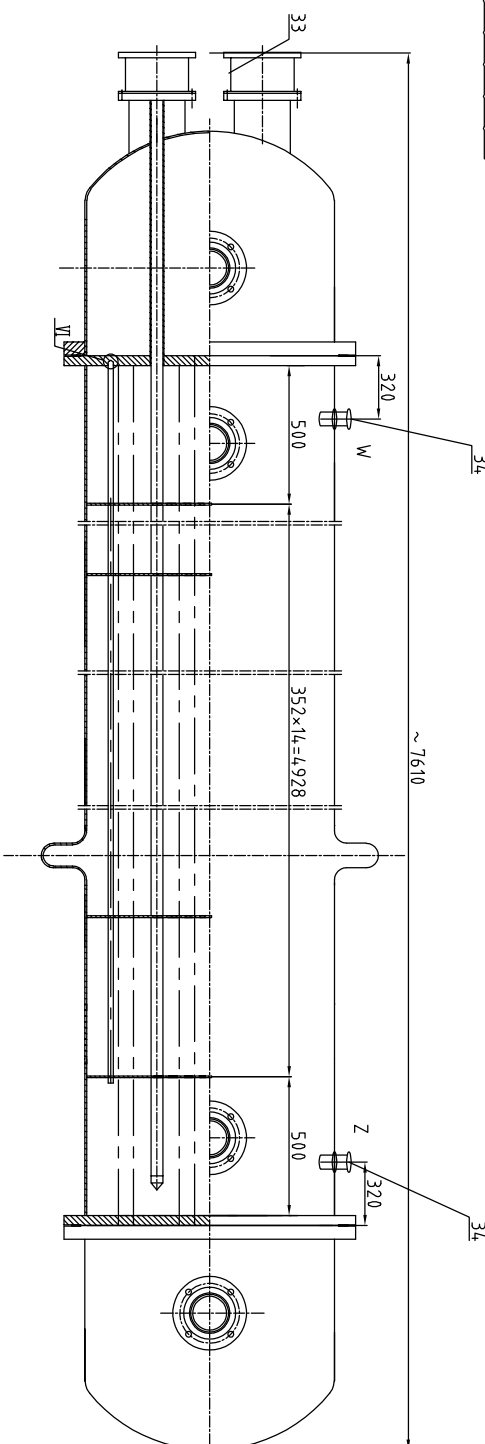
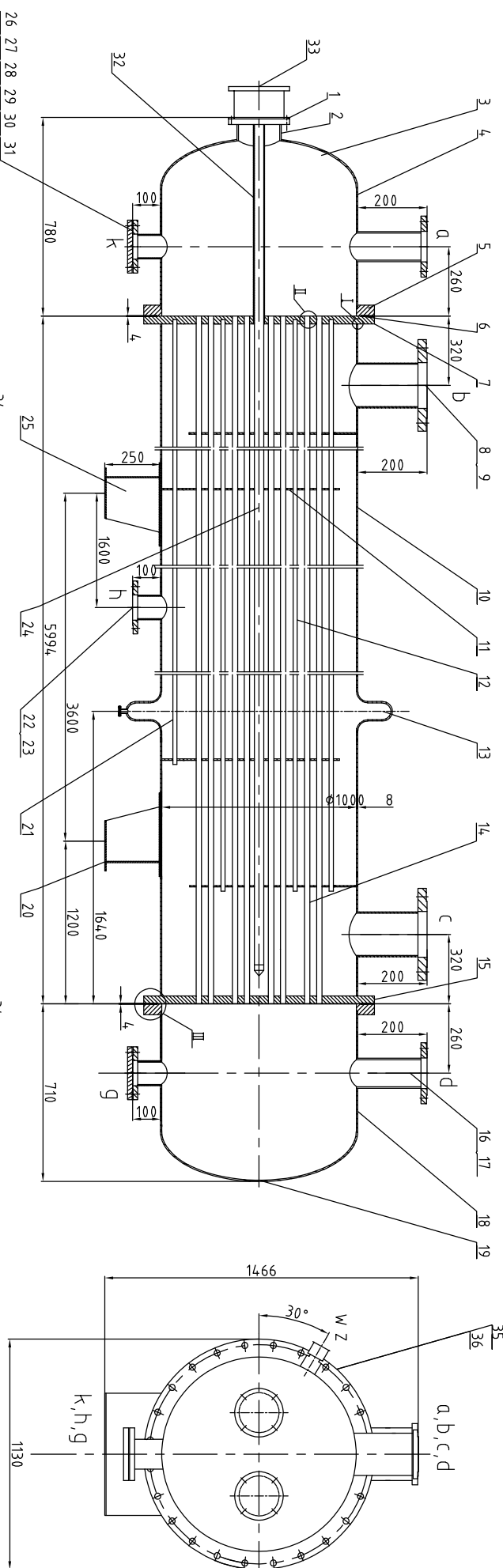
Lanzhou Xingye Material Protection Co., Ltd.

2025.12

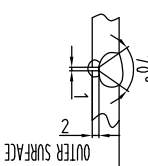
DESIGNED		Anodically Protected	Position	EX-11302
DRAWING		first absorption	number	
CHECKED		Acid Cooler	DRAWING NO.	HR2509-0
REVIEWED				
APPROVED		The 500,000-ton sulfur-based acid production project in Uzkentistan	SCALE	WEIGHT(Kg) 5420

TECHNICAL REQUIREMENTS

1. The design, manufacture, quality control and acceptance of anodic protection acid cooler refer to the standards of GB/T150.1-150.4-2024
2. "Pressure Vessel" and GB/T151-2014, "Heat Exchanger".
3. 2. Anode protection acid cooler manufacturing production management and process control and quality inspection in accordance with the enterprise standard LZZY GL 001-2011 "Production Process Control Procedures",
4. LZZY GL 002-2011 "Acid cooler manufacturing process and Quality Inspection Code"
5. 3. The welding groove type shall be selected in accordance with GB/T 985.1 - 2008 Recommended Grooves for Oxyfuel Gas Welding, Shielded Metal Arc Welding, Gas-Shielded Welding, and High-Energy Beam Welding For groove processing, a mechanical method shall be used for fabrication, or after thermal cutting, it shall be trimmed by grinding wheel matching.
6. 4. The weld surface of the acid cooler must not have cracks, incomplete penetration, lack of fusion, pores, arc craters, slag inclusions, and spatter on both sides of the weld. For stainless steel welded joints, there must be no undercuts.
7. 5. In accordance with GB/T 150.1 - 150.4 - 2024, Pressure Vessel, hydrostatic tests shall be conducted on the tubes and the shell side. The test pressure shall be 1.25 times the design pressure.
8. 6. After the equipment is manufactured, the stainless steel surface shall be subjected to pickling and passivation treatment using a pickling and passivating paste that does not contain hydrochloric acid.
9. 7. Painting, packaging, transportation according to NB/T10558-2021 "Pressure Vessel coating and transportation packaging" provisions.



LONGITUDINAL AND CIRCUMFERENTIAL WELDS OF SHELL



TECHNICAL CHARACTERISTIC TABLE

NAME	UNIT	TUBE SIDE	SHELL SIDE
TYPE OF ASSET		SHELL AND TUBE HEAT EXCHANGER	
DESIGN PRESS	Mpa	0.5	0.6
DESIGN TEMP	°C	75	120
OPERATING PRESS	Mpa	0.4	0.5
OPERATING TEMP	°C	30-75	92.5-71
MEDIUM NAME		Desalted water H2SO4 98.5%	
MEDIUM DENSITY	Kg/h	79000	414800
HEAT EXCHANGER TYPE	M ²	265	
HEAT TRANSFER COEFFICIENT	W/(m ² °C)	407	
INLET EFFICIENCY		0.85	0.85
OPERATING MEDIA	kg	19900	

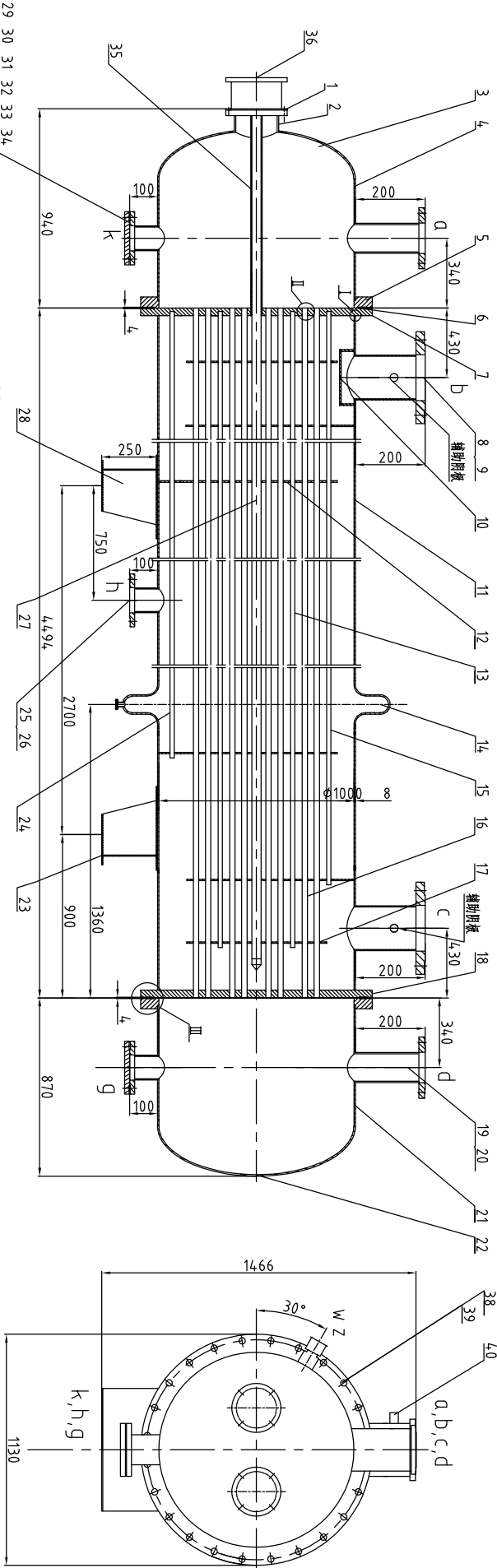
NOZZLE SCHEDULE

MARK	NOMINAL SIZE	CONNECTION STANDARD	CONNECTION SURFACE FORM	USER OR NAME
a	150	PL150(B)-10.HG/T20592	RF	Water Outlet
b	300	SO300(B)-16.HG/T20592	RF	Acid Inlet
c	300	SO300(B)-16.HG/T20592	RF	Acid Outlet
d	150	PL150(B)-10.HG/T20592	RF	Water Inlet
g	80	PL80(B)-10.HG/T20592	RF	Sewage outlets
h	80	SO80(B)-16.HG/T20592	RF	Sewage outlets
k	80	PL80(B)-10.HG/T20592	RF	Sewage outlets

36	GB6170	NUTS M20	Q235B	72	
35	GB5782	BOLTS M20x150	Q235B	72	
34		REFERENCE ELECTRODE	ASSEMBLY	2	
33		MAIN CATHODE	ASSEMBLY	2	
32		CATHODE CASING Ø40x3.5L=805	316L	2	
31	GB6170	NUTS M16	Q235B	16	
30	GB5782	BOLTS M16x65	Q235B	16	
29	HG/T20592	FLANGE COVER BL80-10	Q235B	2	
28	HG/T20592	GASKETS RF80-10, Ø=2	RUBBER SHEET	2	
27	HG/T20592	PIPE FLANGE PL80(B)-10-RF	Q235B	2	
26		NOZZLE Ø89x4	20	2	
25	JB/T4712.1-2007	LEFT SUPPORT BL1000-F,H=250	Q235B	1	
24		CATHODE FRONT Ø20L=6585XY-1 ALLOY	2		
23	HG/T20592-2009	PIPE FLANGE SO80(B)-16-RF	304L	1	
22		NOZZLE Ø89x4	304L	1	
21		hot pull rod Ø14,L=5376	304L	1	
20	JB/T4712.1-2007	RIGHT SUPPORT BL1000-S,H=250	Q235B	1	
19	GB/T25198-2010	ELLIPTICAL HEAD DN1000×8	Q235B	1	
18		RIGHT CHANNEL NOZZLE STUD DN1000x4L=394	Q235B	1	
17	HG/T20592-2009	PIPE FLANGE PL150(B)-10-RF	Q235B	2	
16		NOZZLE Ø159x5	20	2	
15	HR2510-3	RIGHT TUBE SHEET Ø=33	304L	1	
14		HEAT EXCHANGER TUBES Ø19x1.5L=4500	316L	74.9	
13	GB/T 1674.9	EXPANSION JOINT DN1000,Ø=8	304L	1	
12		Long pull rod Ø14,L=5728	304L	9	
11	HR2510-2	BAFFLE Ø=8	304L	15	
10		SHELL-SIDE CYLINDER DN1000x8L=5928	304L	1	
9	HG/T20592-2009	PIPE FLANGESO30(B)-16-RF	304L	2	
8		NOZZLE Ø325x8	304L	2	
7	HR2510-1	LEFT TUBE SHEET Ø=33	304L	1	
6		GASKETS DN1000 PNO.6,Ø=4	RUBBER SHEET	2	
5	NB/T 47021	FLANGE DN1000 PNO.6	Q235B	2	
4		LEFT CHANNEL NOZZLE STUD DN1000x4L=394	Q235B	1	
3	GB/T25198-2010	ELLIPTICAL HEAD DN1000×8	Q235B	1	
2		NOZZLE Ø159x5	20	2	
1		CATHODE HOLDER Ø220,δ=20	Q235B	2	
PART NO.	DWG OR STANDARD NO.	NAME	MATERIAL	QT	UNIT
					TOT WT (KG)
Lanzhou Xingye Material Protection Co., Ltd.					REMARKS
					2025.12
DESIGNED		Desalted water		Position number	EX-11303
DRAWING		preheater		DRAWING NO.	HR2510-0
CHECKED					
REVIEWED		The 500,000-ton sulfur-based acid production project in Uzbekistan		SCALE	WEIGHT(kg) 6000
APPROVED					

管口表

符号	工程尺寸	链接尺寸标准	连接面形式	用途或名称
a	300	PL300(B)-10,HG/T20592	RF	出水口
b	500	SO500(B)-16,HG/T20592	RF	进酸口
c	500	SO500(B)-16,HG/T20592	RF	出酸口
d	300	PL300(B)-10,HG/T20592	RF	进水口
g	80	PL80(B)-10,HG/T20592	RF	排污口
h	80	SO80(B)-16,HG/T20592	RF	排污口
k	80	PL80(B)-10,HG/T20592	RF	排污口

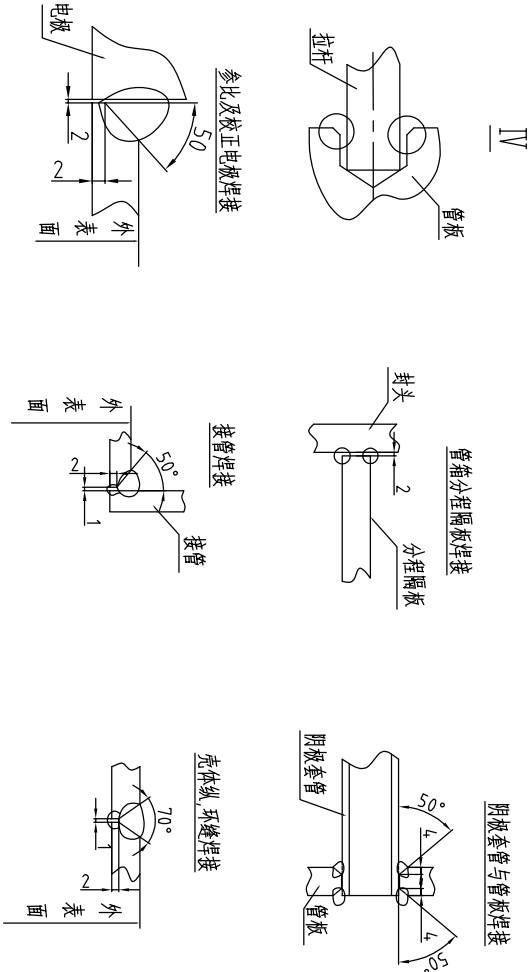
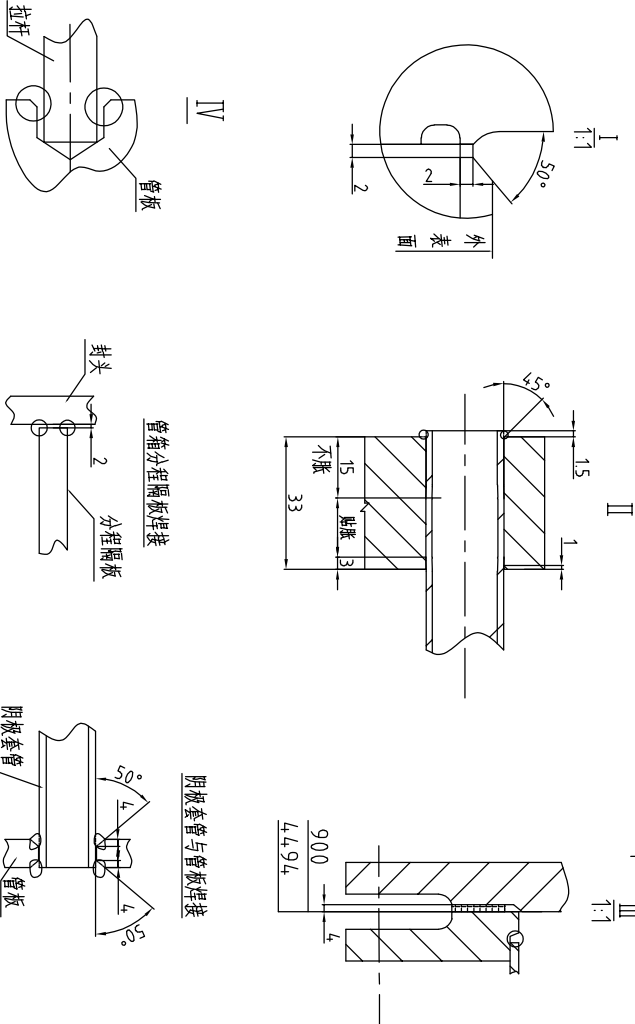


技术特性表

名称	单位	管程	壳程
容器类别		管壳式换热器	
设计压力	Mpa	0.5	0.6
设计温度	℃	75	120
工作压力	Mpa	0.4	0.5
工作温度	℃	32-4.0	88.5-80
物料名称		循环水	硫酸 98.5%
物料流量	Kg/h	422271	1066800
换热面积	M²		105
传热系数	W/(m²℃)		667.1
焊缝系数		0.85	0.85
操作重量	kg		17800

技术要求

1. 阳极保护酸冷器设计、制造、质量控制和验收参照GB/T150.1~150.4-2024《压力容器》和GB/T151-2014《热交换器》标准。
2. 阳极保护酸冷器制造生产管理和工艺控制以及质量检验遵照企业标准LZXYY.GL.001-2011《生产过程控制程序》、LZXYY.GL.002-2011《酸冷器制造工艺和质量检验守则》。
3. 焊接坡口型式依据GB/T985.1-2008《气焊、焊条电弧焊、气体保护焊和高能束焊的推荐坡口》选取，坡口加工采用机械方法制作或热切割后再进行砂轮磨加工修整。
4. 酸冷器焊缝表面不得有裂纹、未焊透、未熔合、气孔、弧坑、夹渣以及焊缝两侧飞溅等；不锈钢焊接接头不得有咬边。
5. 依据GB/T150.1~150.4-2024《压力容器》对管、壳程进行水压试验，试验压力为1.25倍的设计压力。
6. 设备制造完毕后，不锈钢表面使用不含盐酸的酸洗钝化膏进行酸洗钝化处理。
7. 油漆、包装、运输按NB/T10558-2021《压力容器涂敷与运输包装》的规定。



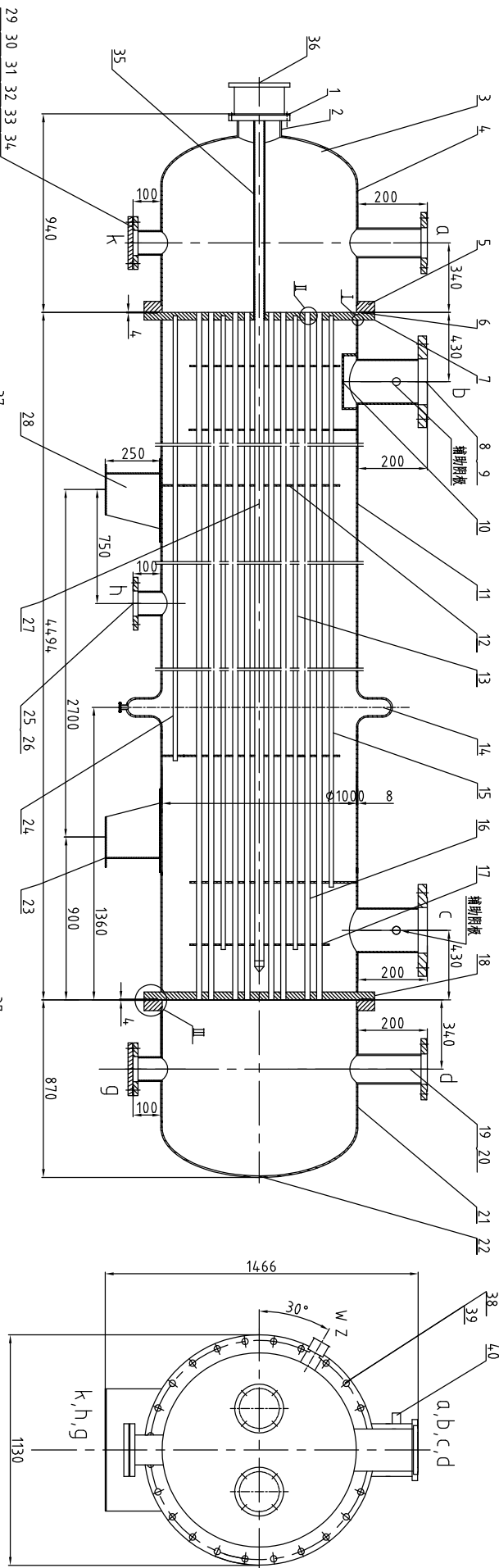
设计				位号	
制图				EX-11304	
审核				HR2511-0	
审定				重量Kg	
二吸酸冷却器				4105	

兰州兴业材料保护有限公司

2025年12月

NOZZLE SCHEDULE

MARK	NOMINAL SIZE	CONNECTION STANDARD	DIMENSION SURFACE FORM	USER OR NAME
a	300	PL300(B)-10-HG/T20592	RF	Water Outlet
b	500	S0500(B)-16-HG/T20592	RF	Acid Inlet
c	500	S0500(B)-16-HG/T20592	RF	Acid Outlet
d	300	PL300(B)-10-HG/T20592	RF	Water Inlet
g	80	PL80(B)-10-HG/T20592	RF	Sewage outlets
h	80	S080(B)-16-HG/T20592	RF	Sewage outlets
k	80	PL80(B)-10-HG/T20592	RF	Sewage outlets

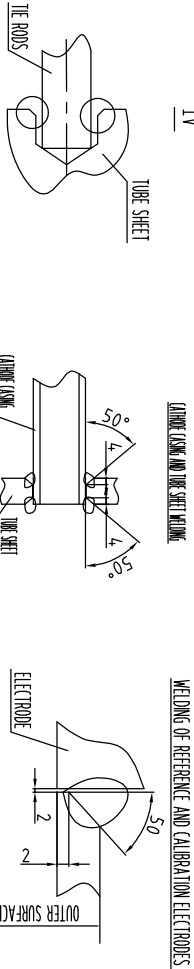
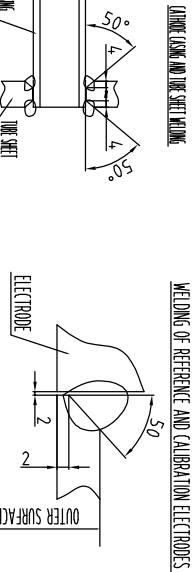
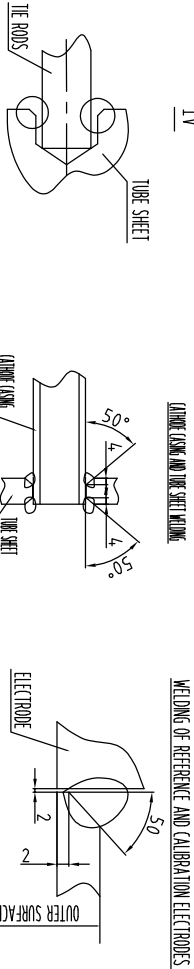
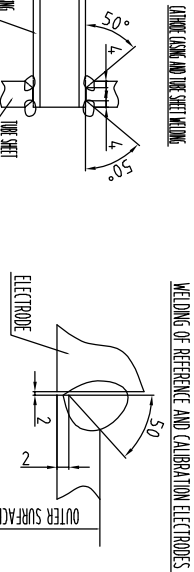
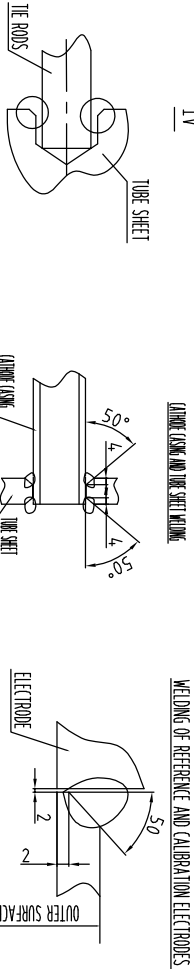
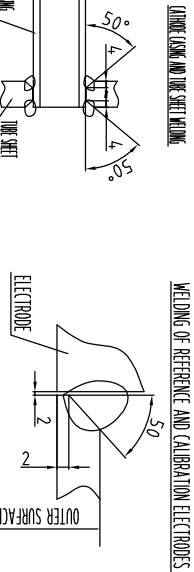
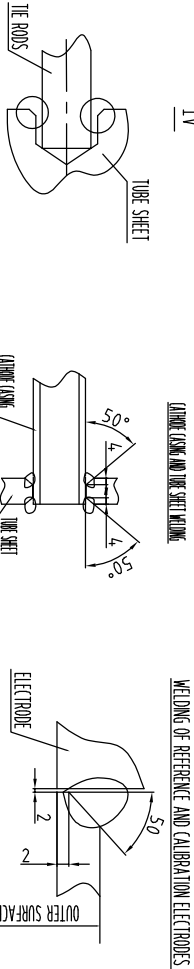
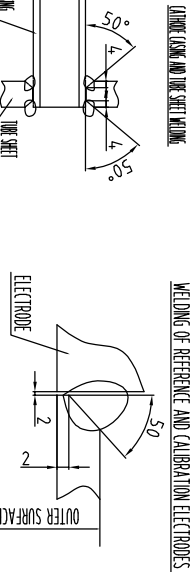
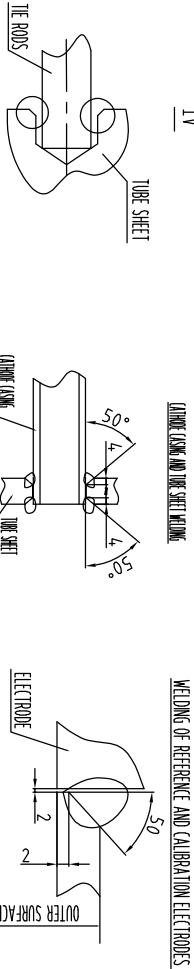
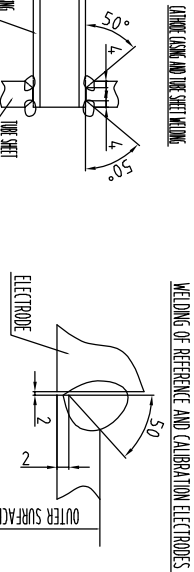
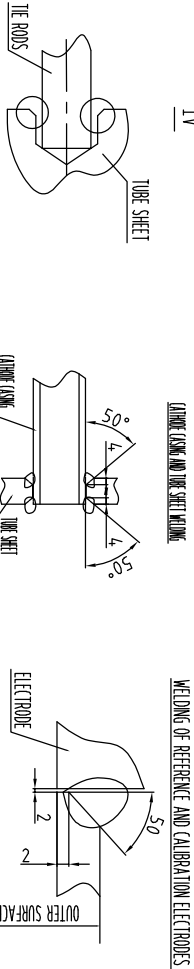
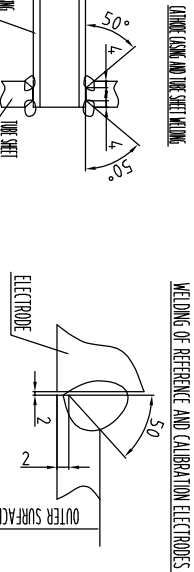
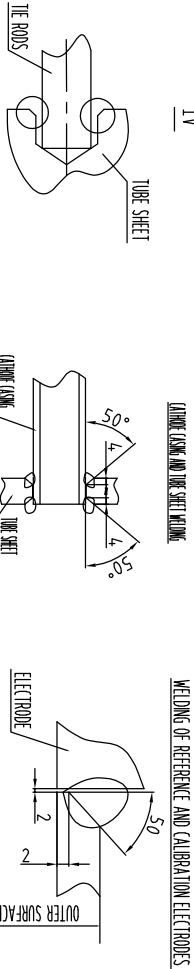
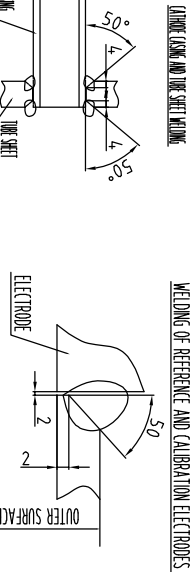
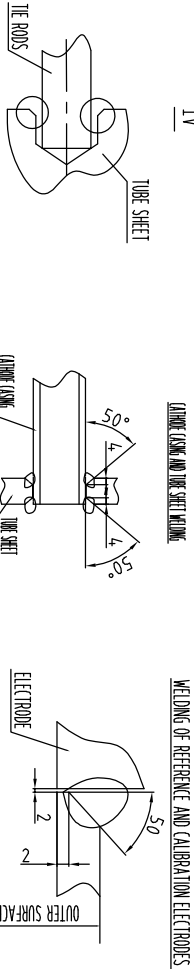
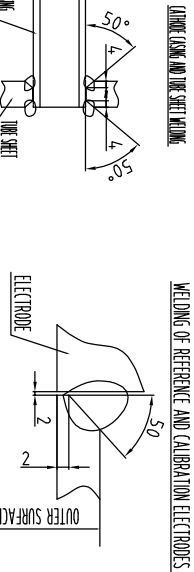
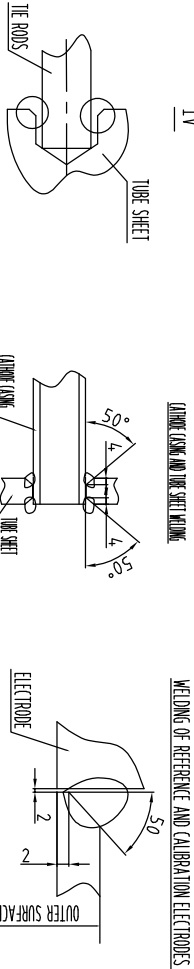
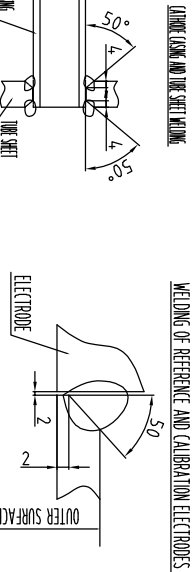
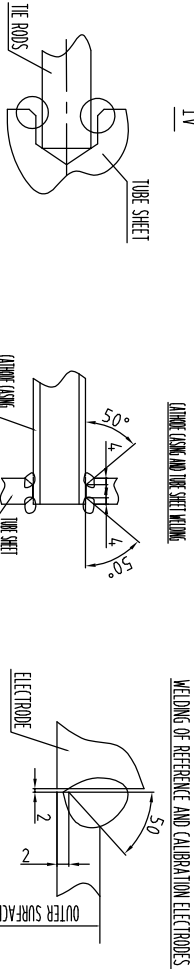
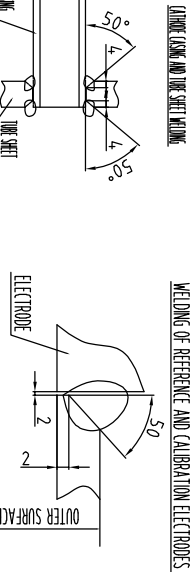
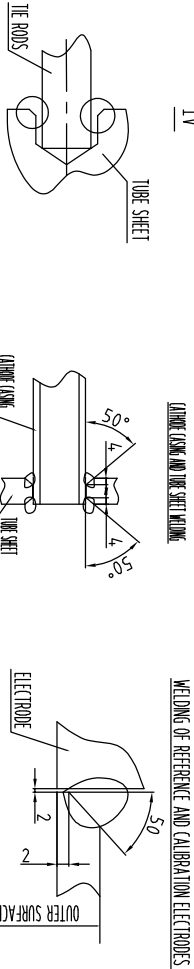
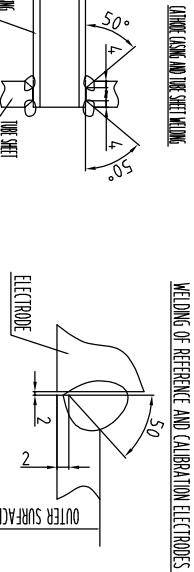
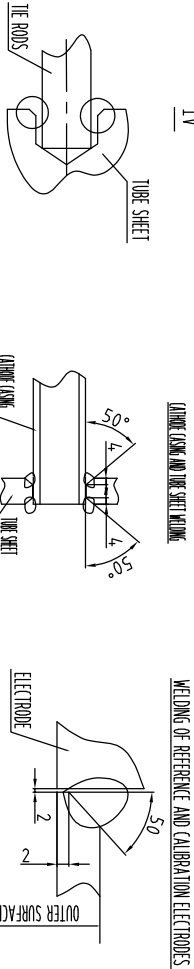
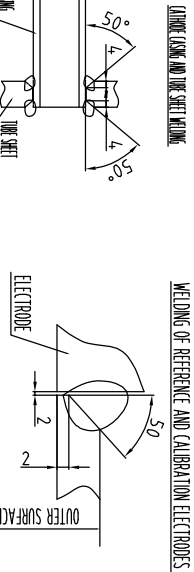
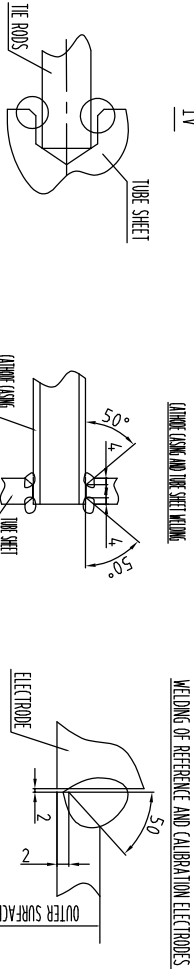
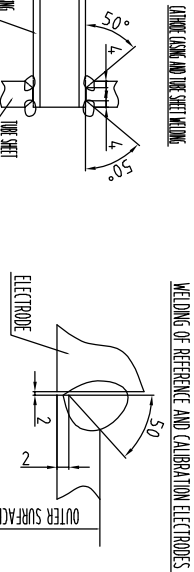
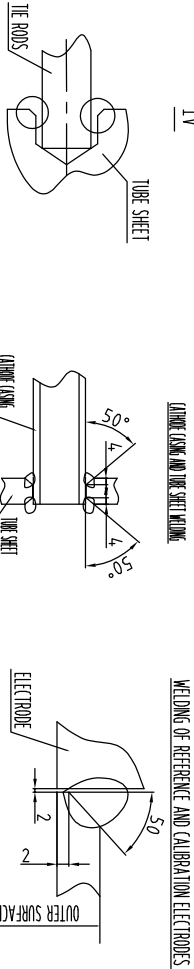
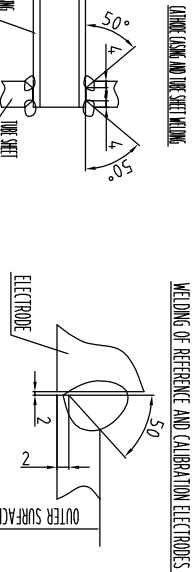
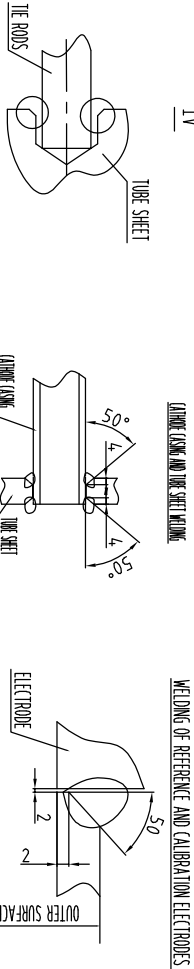
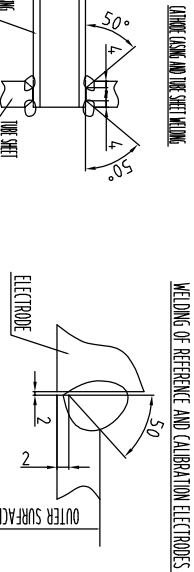
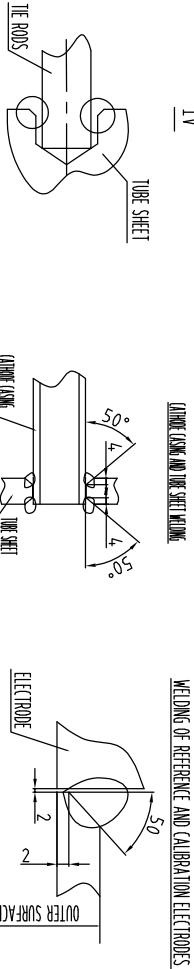
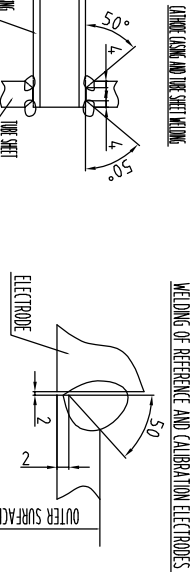
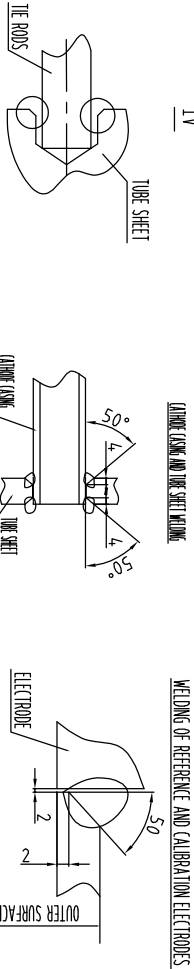
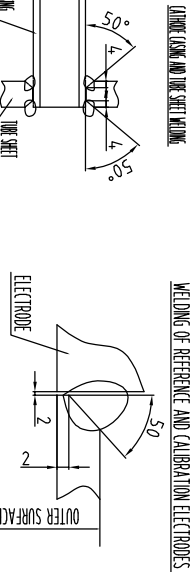
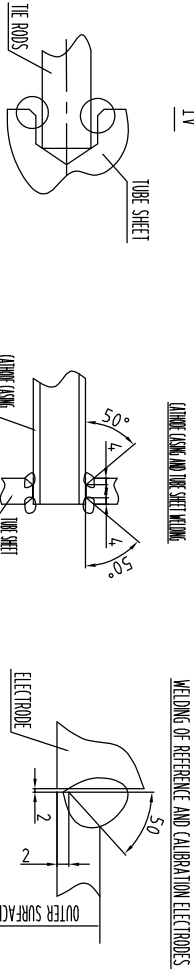
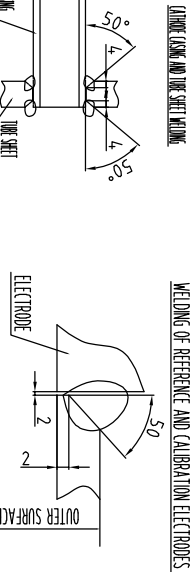
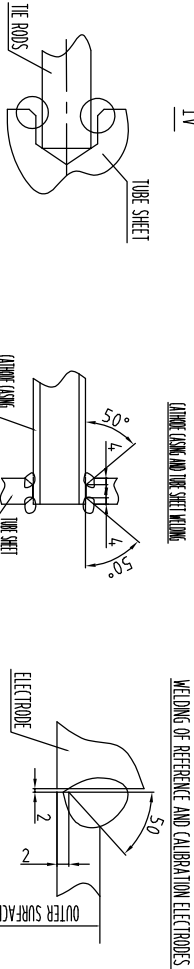
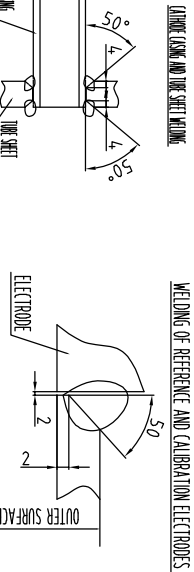
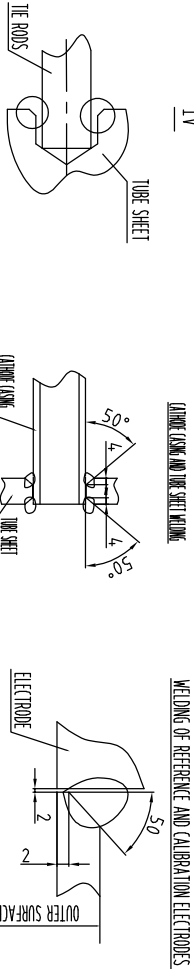
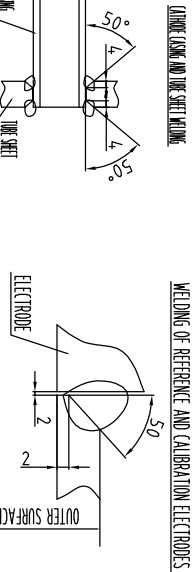
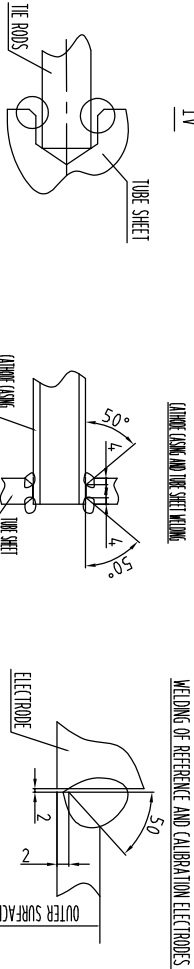
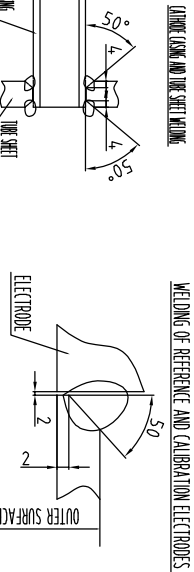
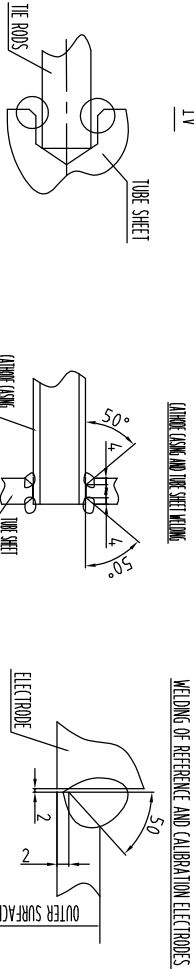
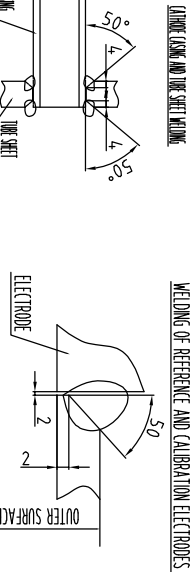
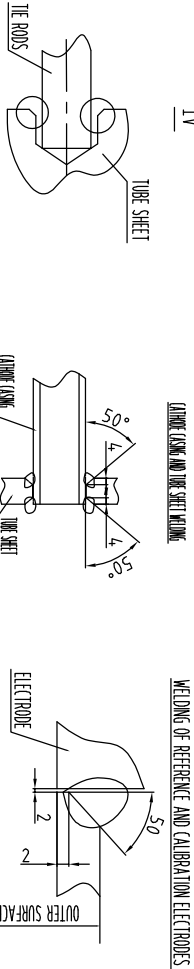
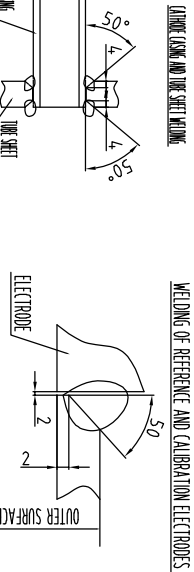
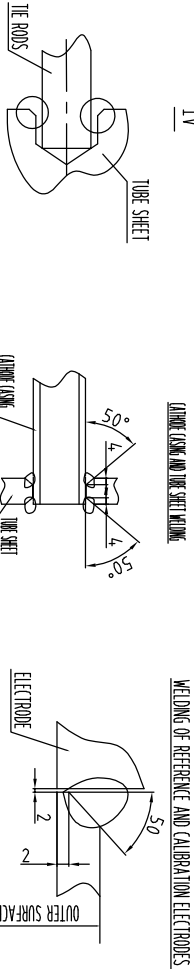
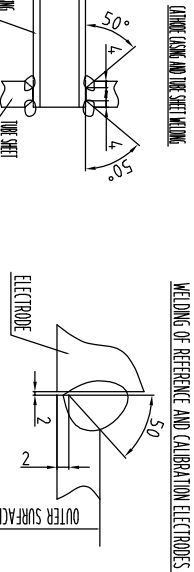
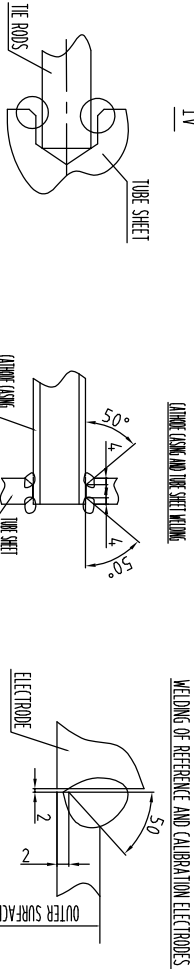
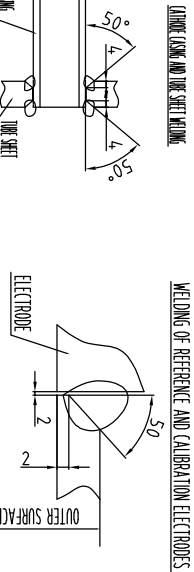
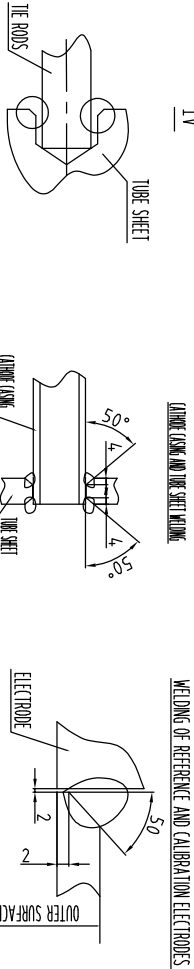
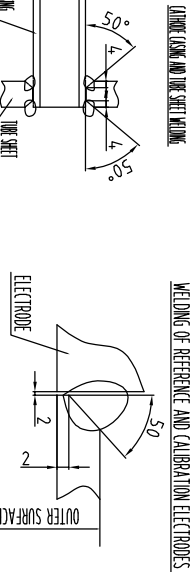
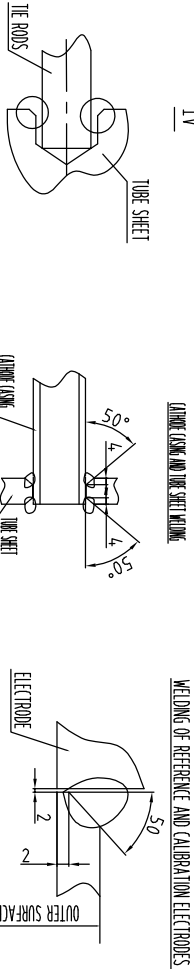
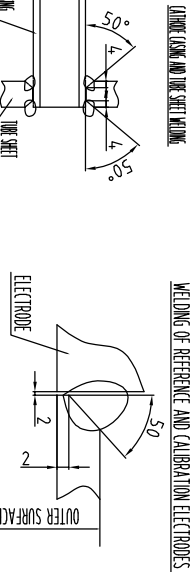
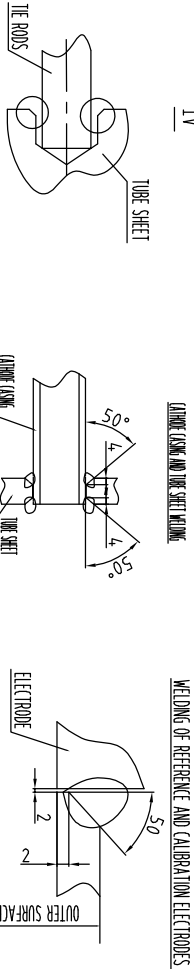
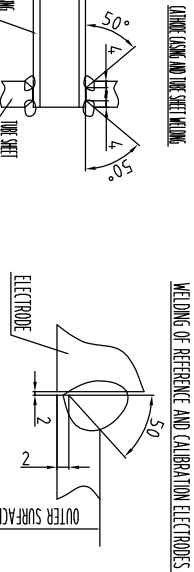
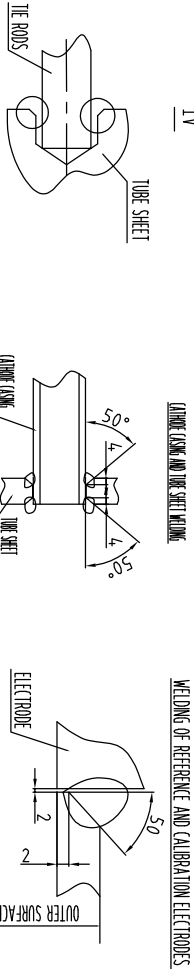
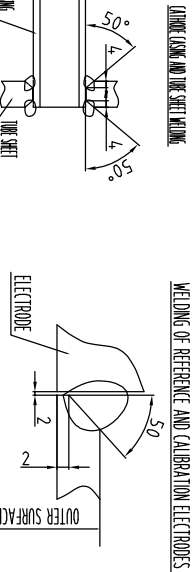
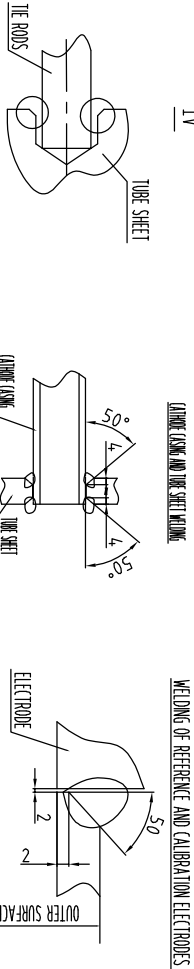
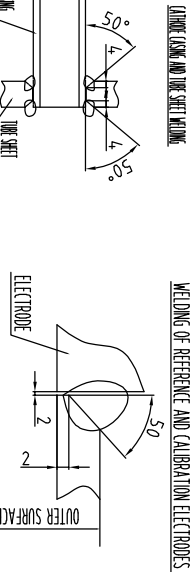
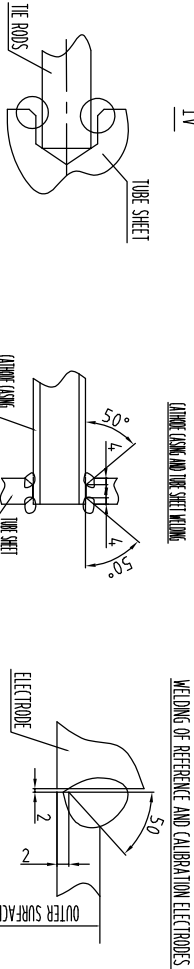
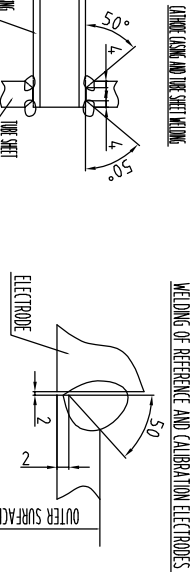
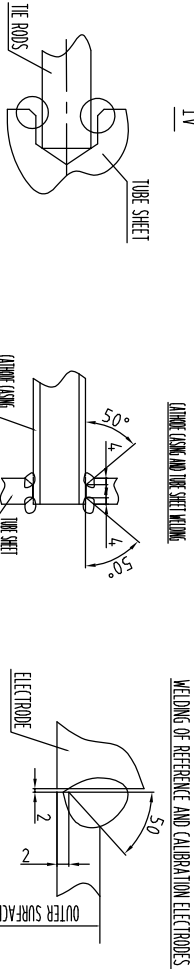
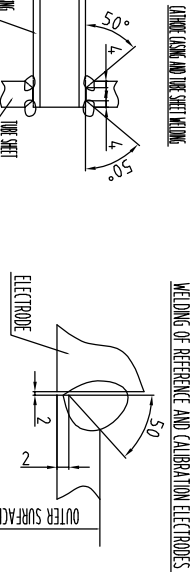
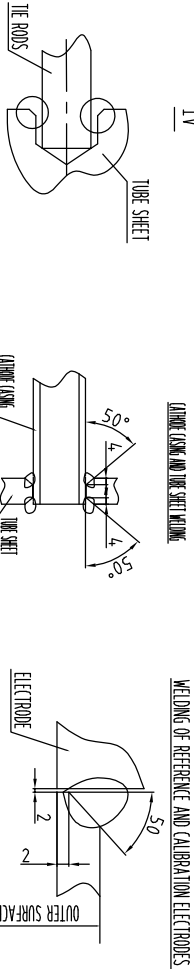
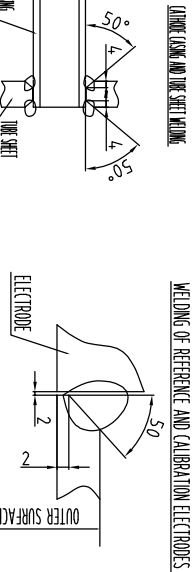
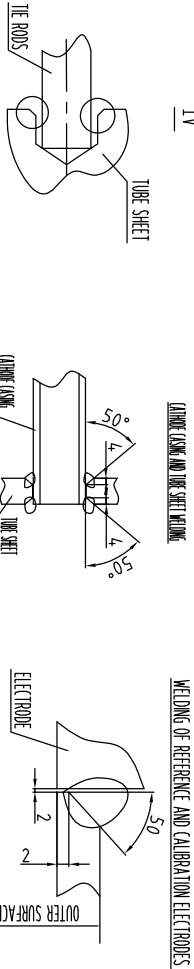
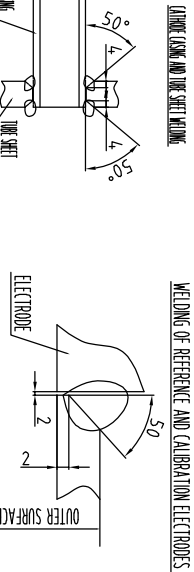
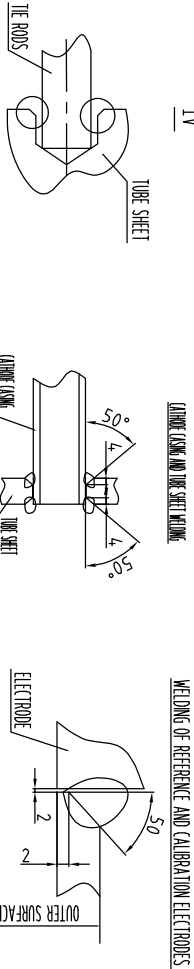
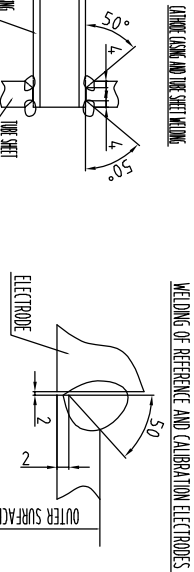
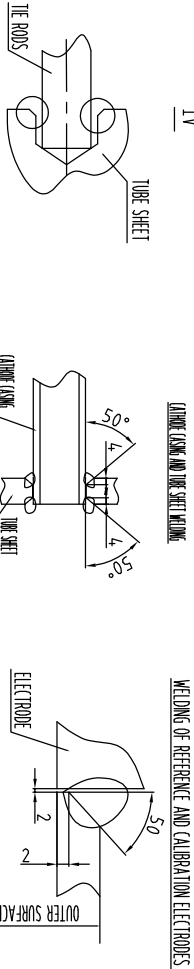
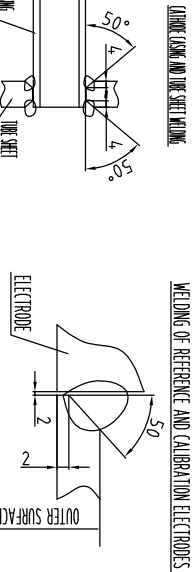
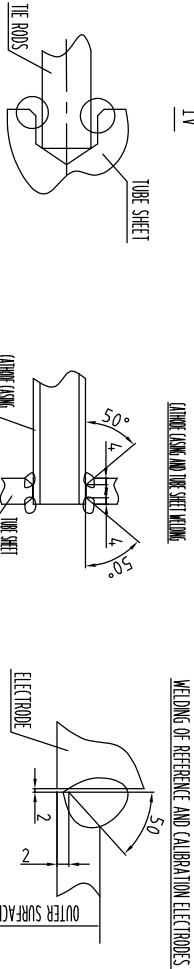
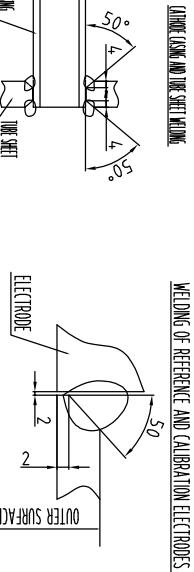
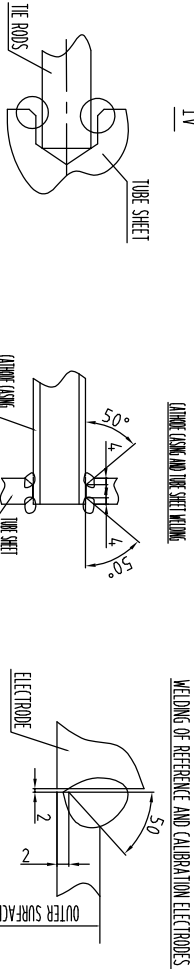
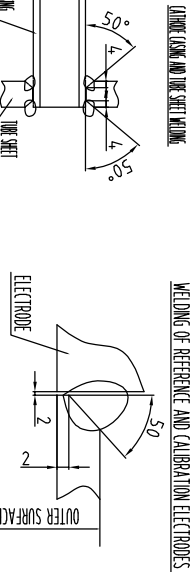
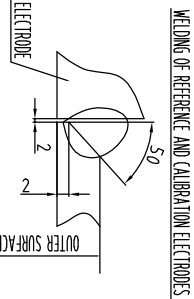
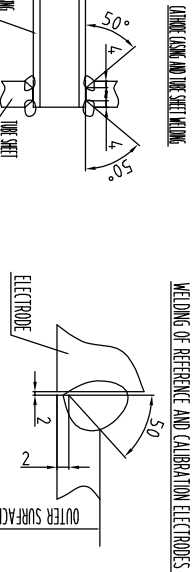
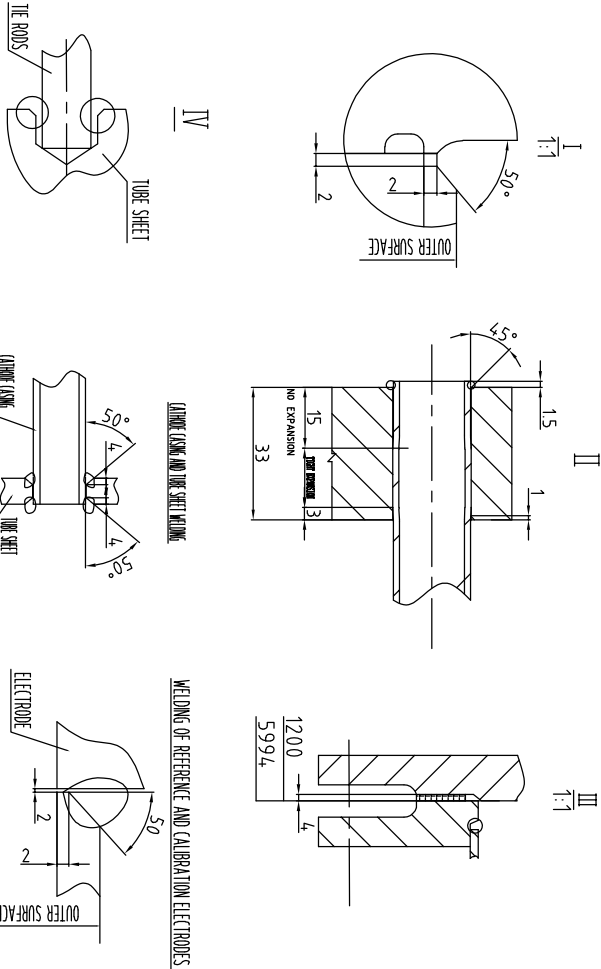


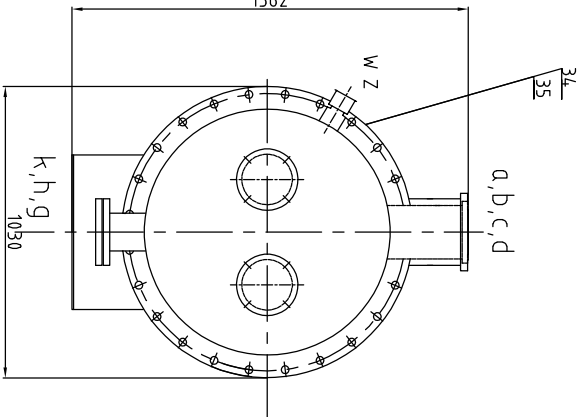
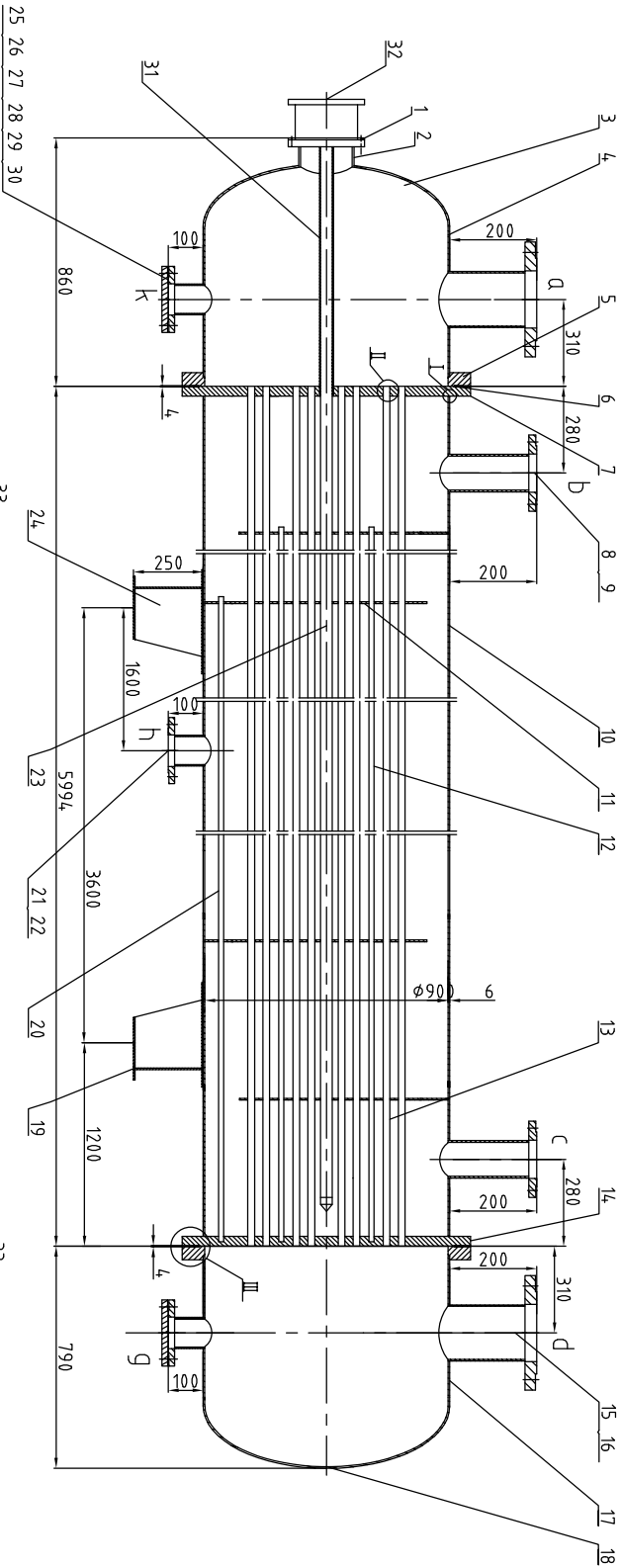
TECHNICAL CHARACTERISTIC TABLE

NAME	UNIT	TUBE SIDE	SHELL SIDE
TYPE OF VESSEL		SHELL AND TUBE HEAT EXCHANGER	
DESIGN PRESS	Mpa	0.5	0.6
DESIGN TEMP	℃	75	120
OPERATING PRESS	Mpa	0.4	0.5
OPERATING TEMP	℃	32-40	88.5-80
MEDIUM NAME		Circulation Water H2SO4 98.5%	
MEDIUM DISCHARGE	Kg/h	4.22271	1086800
HEAT EXCHANGE AREA	M ²	105	
HEAT TRANSFER COEFFICIENT	W/(m ² ·℃)	667.1	
OVERALL EFFICIENCY		0.85	0.85
OPERATING WEIGHT	kg		17800

TECHNICAL REQUIREMENTS

- 1.The design, manufacture, quality control and acceptance of anodic protection acid cooler refer to the standards of GB/T150.1-150.4-2024 “Pressure Vessel” and GB/T151-2014, “Heat Exchanger”.
- 2.Anode protection acid cooler manufacturing production management and process control and quality inspection in accordance with the enterprise standard LZXY.GL.001-2011 “Production Process Control Procedures”, LZXY.GL.002-2011 “Acid cooler manufacturing process and Quality Inspection Code” LZXY.GL.001-2011 “Production Process Control Procedures”, Recommended grooves for Oxyfuel Gas Welding, Shielded Metal Arc Welding, Gas-Shielded Welding, and High-Energy Beam Welding. For groove processing, a mechanical method shall be used for fabrication,or after thermal cutting, it shall be trimmed by grinding wheel machining .
4. The weld surface of the acid cooler must not have cracks, incomplete penetration, lack of fusion, pores, arc craters, slag inclusions, and spatter on both sides of the weld. For stainless steel welded joints, there must be no undercuts.
5. In accordance with GB/T 150.1 - 150.4 - 2024 Pressure Vessel hydrostatic tests shall be conducted on the tubes and the shell side. The test pressure shall be 1.25 times the design pressure.
6. After the equipment is manufactured, the stainless steel surface shall be subjected to pickling and passivation treatment using a pickling and passivating paste that does not contain hydrochloric acid.
- 7.Painting, packaging, transportation according to NB/T10558-2021 “Pressure Vessel coating and transportation packaging” provisions.

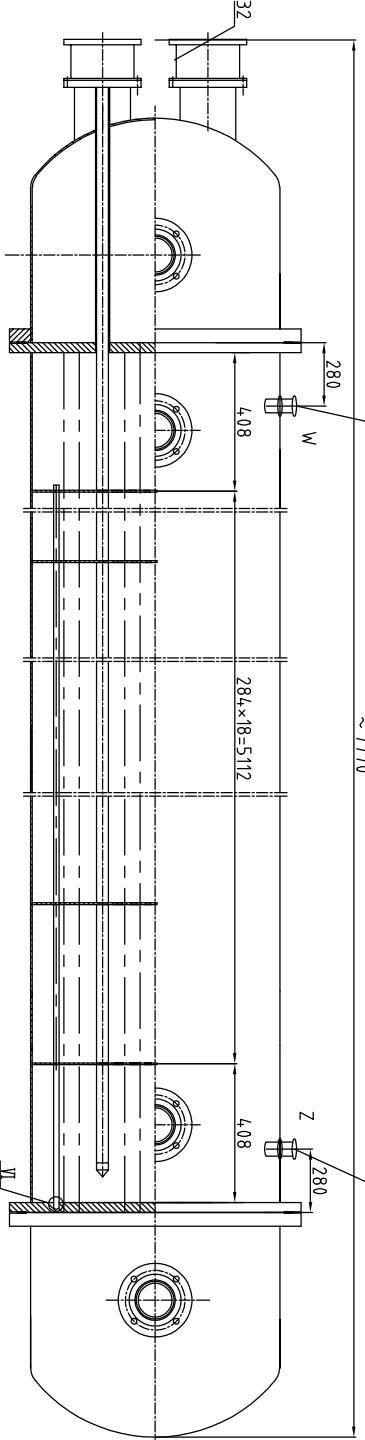




管口表				
符号	工程尺寸	链接尺寸标准	连接面形式	用途或名称
a	250	PL250(B)-10,HG/T20592	RF	出水口
b	200	SO200(B)-16,HG/T20592	RF	进酸口
c	200	SO200(B)-16,HG/T20592	RF	出酸口
d	250	PL250(B)-10,HG/T20592	RF	进水口
g	80	PL80(B)-10,HG/T20592	RF	排污口
h	80	SO80(B)-16,HG/T20592	RF	排污口
k	80	PL80(B)-10,HG/T20592	RF	排污口

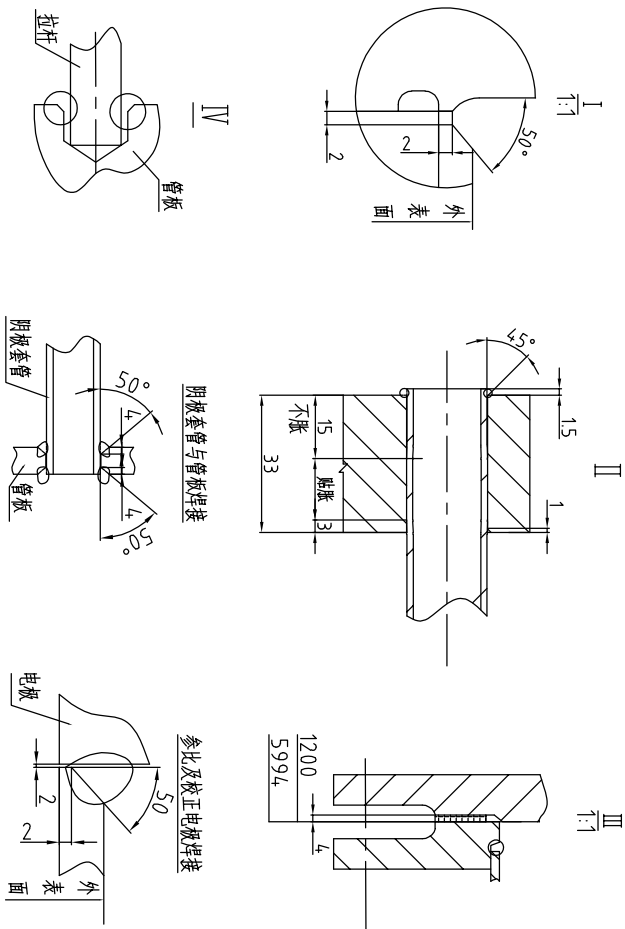
技术特性表

名称	单位	管程	壳程
设备类别		管壳式换热器	
设计压力	Mpa	0.5	0.6
设计温度	℃	75	120
工作压力	Mpa	0.4	0.5
工作温度	℃	32-4.0	70.5-4.0
物料名称	Kg/h	循环水	硫酸 93%
物料用量		276859	187200
换热面积	M ²	295	
传热系数	W/(m ² ·℃)	4464	
焊接系数		0.85	0.85
操作重量	kg	15900	

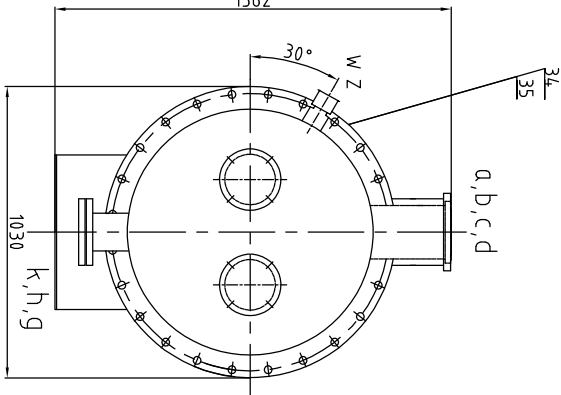
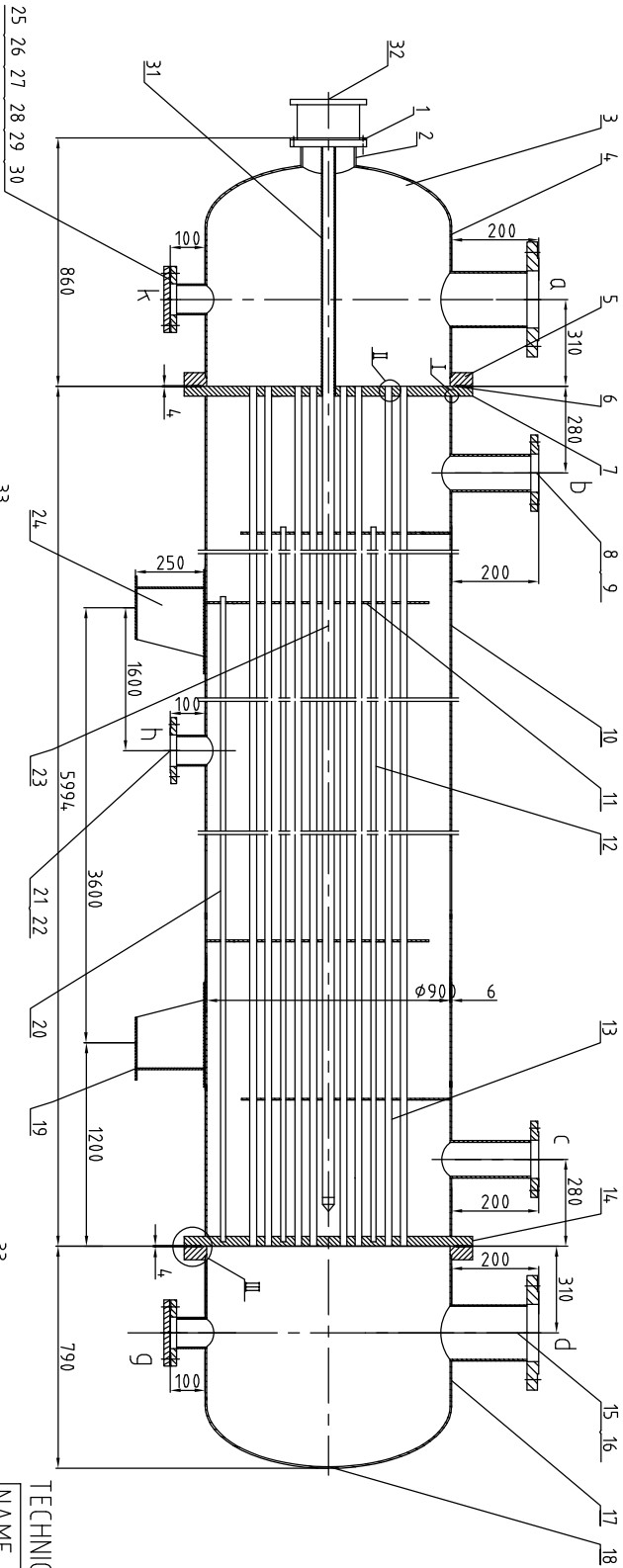


技术要求

- 阳极保护酸冷器设计、制造、质量控制和验收参照GB/T150.1~150.4-2024《压力容器》和GB/T151-2014《热交换器》标准。
- 阳极保护酸冷器制造生产管理和工艺控制以及质量检验遵照企业标准LZX.Y.GL.002-2011《酸冷器制造工艺和质量检验守则》。
- 焊接坡口型式依据GB/T985.1-2008《气焊、焊条电弧焊、气体保护焊和高能束焊的推荐坡口》选取，坡口加工采用机械方法制作或热切割后再进行砂轮磨加工修整。
- 酸冷器焊缝表面不得有裂纹、未焊透、未熔合、气孔、弧坑、夹渣以及焊缝两侧飞溅等；不锈钢焊接接头不得有咬边。
- 依据GB/T150.1~150.4-2024《压力容器》对管、壳程进行水压试验，试验压力为1.25倍的设计压力。
- 设备制造完毕后，不锈钢表面使用不含盐酸的酸洗钝化膏进行酸洗钝化处理。
- 油漆、包装、运输按NB/T10558-2021《压力容器涂敷与运输包装》的规定。



兰州兴业材料保护有限公司				2025年12月	
设计				位号	EX-11305
制图				图号	HR2512-0
审核					
审定				马兹别克斯坦年产50万吨硫酸制酸项目	重量kg 5585

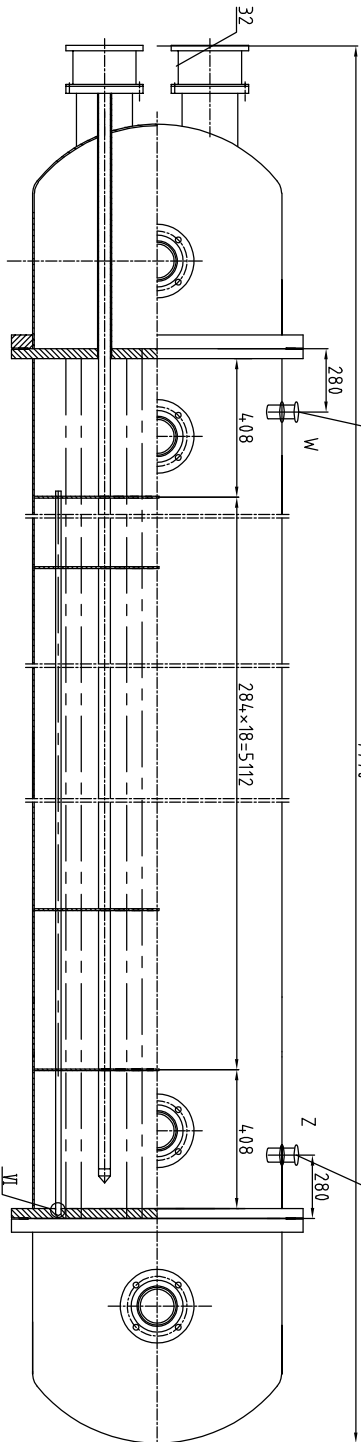


MARK	NOMINAL SIZE	CONNECTION STANDARD	CONNECTION SURFACE FORM	USER OR NAME
a	250	PL250(B)-10.HG/120592	RF	Water Outlet
b	200	SO200(B)-16.HG/120592	RF	Acid Inlet
c	200	SO200(B)-16.HG/120592	RF	Acid Outlet
d	250	PL250(B)-10.HG/120592	RF	Water Inlet
g	80	PL80(B)-10.HG/120592	RF	Sewage outlets
h	80	SO80(B)-16.HG/120592	RF	Sewage outlets
k	80	PL80(B)-10.HG/120592	RF	Sewage outlets

NOZZLE SCHEDULE

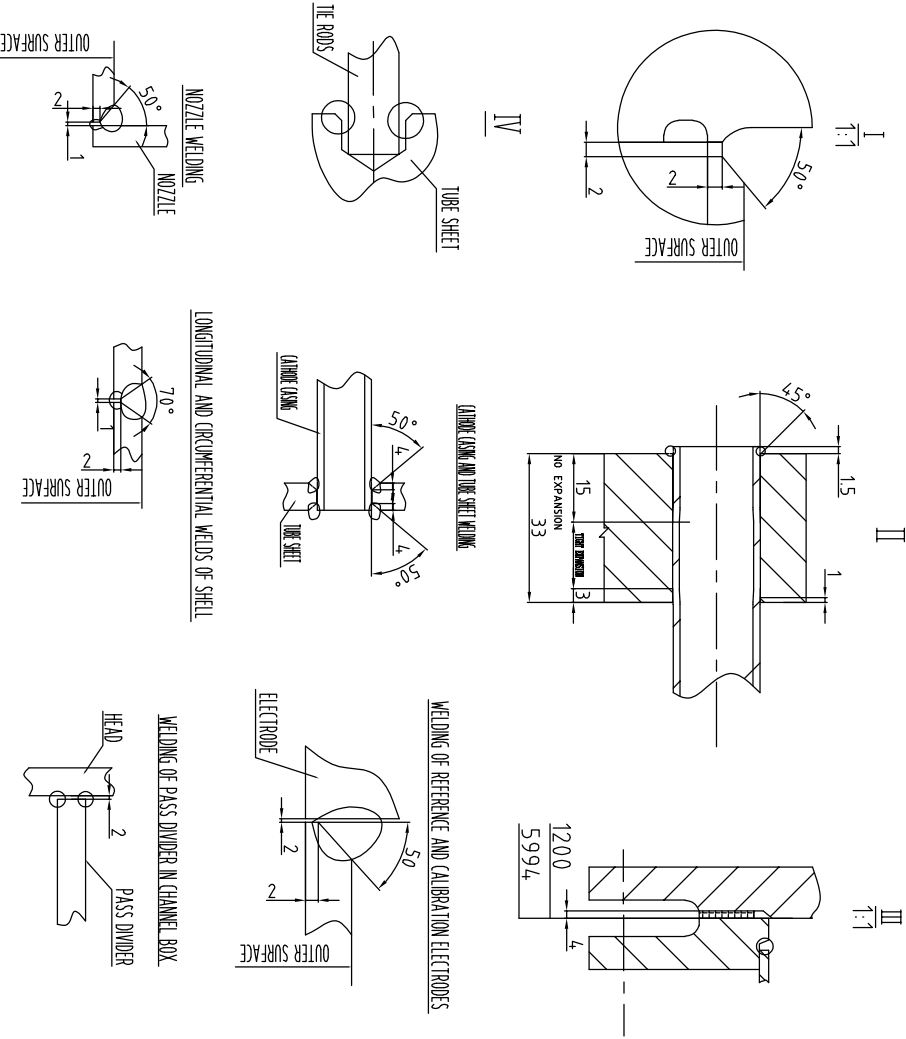
TECHNICAL CHARACTERISTIC TABLE

NAME	UNIT	TUBE SIDE	SHELL SIDE
TYPE OF VESSEL		SHELL AND TUBE HEAT EXCHANGER	
DESIGN PRESS	Mpa	0.5	0.6
DESIGN TEMP	℃	75	120
OPERATING PRESS	Mpa	0.4	0.5
OPERATING TEMP	℃	32-4.0	70.5-4.0
FLOW RATE	Kg/h	Circulation Water	H2SO4 93%
HEATING DISAP	M ²	276859	187200
HEAT EXCHANGER	W/(m ² ℃)	295	446.4
MAX EFFICIENCY		0.85	0.85
OPERATING WEIGHT	kg	15900	



TECHNICAL REQUIREMENTS

- The design, manufacture, quality control and acceptance of anodic protection acid cooler refer to the standards of GB/T150.1-150.4-2024 “Pressure Vessel” and GB/T151-2014, “Heat Exchanger”.
- Anode protection acid cooler manufacturing production management and process control and quality inspection in accordance with the enterprise standard LZXY.GL.002-2011 “Production Process Control Procedures”.
- The welding groove type shall be selected in accordance with GB/T 985.1 - 2008 Recommended Grooves for Oxyfuel Gas Welding, Shielded Metal Arc Welding, Gas-Shielded Welding, and High-Energy Beam Welding. For groove processing, a mechanical method shall be used for fabrication, or after thermal cutting, it shall be trimmed by grinding wheel machining.
- The weld surface of the acid cooler must not have cracks, incomplete penetration, lack of fusion, pores, arc craters, slag inclusions, and spatter on both sides of the weld. For stainless steel welded joints, there must be no undercut.
- In accordance with GB/T 150.1 - 150.4 - 2024 Pressure Vessel hydrostatic tests shall be conducted on the tubes and the shell side. The test pressure shall be 1.25 times the design pressure.
- After the equipment is manufactured, the stainless steel surface shall be subjected to pickling and passivation treatment using a pickling and passivating paste that does not contain hydrochloric acid.
- Painting, packaging, transportation according to NB/T10558-2021 “Pressure Vessel coating and transportation packaging” provisions.



DESIGNED		Anodically Protected finished acid cooler		Position number	EX-11305
DRAWING		Protected finished acid cooler		DRAWING NO.	HR2512-0
CHECKED					
REVIEWED					
APPROVED		The 500,000-ton sulfur-based acid production project in Uzbekistan		SCALE	WEIGHT(kg) 5585

Lanzhou Xingye Materials Protection Co., Ltd. 2025.12